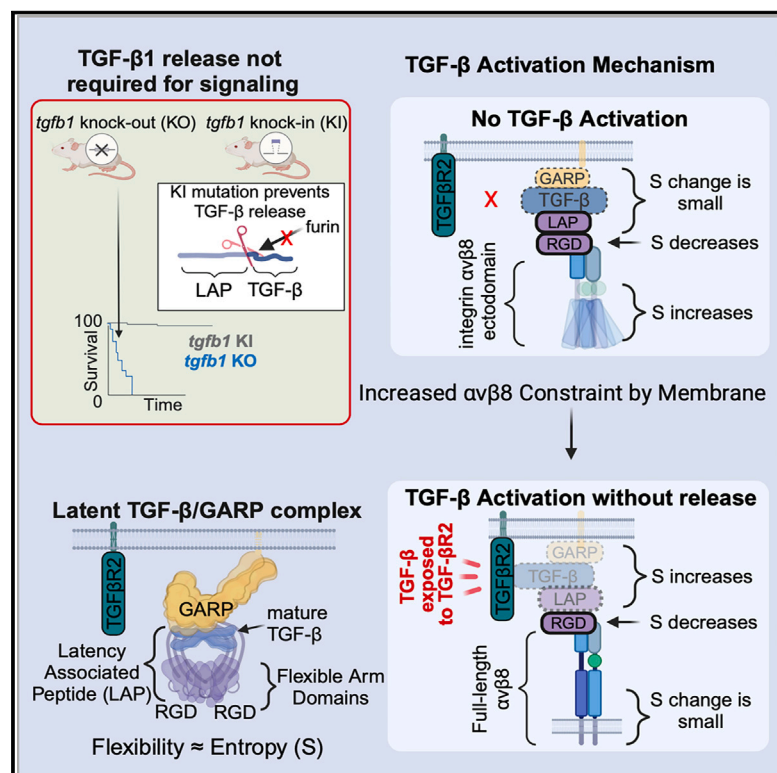


# Dynamic allostery drives autocrine and paracrine TGF- $\beta$ signaling

## Graphical abstract



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## In brief

Genetically engineered mice survive with only autocrine but no paracrine TGF- $\beta$ 1 signaling. Structural and functional studies reveal a mechanism for TGF- $\beta$ 1 autocrine signaling driven by conformational entropy redistribution from  $\alpha$ v $\beta$ 8 binding to the latent TGF- $\beta$ 1/GARP complex. Integrin-mediated entropy redistribution also underlies TGF- $\beta$ 3 activation, suggesting a general mechanism of cell-cell communication.

## Highlights

- Mice survive with only autocrine but no paracrine TGF- $\beta$ 1 signaling
- $\alpha$ v $\beta$ 8 binding activates L-TGF- $\beta$ 1 for autocrine signaling without release
- Conformational entropy redistribution drives allosteric activation of L-TGF- $\beta$  by  $\alpha$ v $\beta$ 8
- Direction of entropy redistribution can be manipulated by stabilizing flexible domains



Article

# Dynamic allostery drives autocrine and paracrine TGF- $\beta$ signaling

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## SUMMARY

TGF- $\beta$ , essential for development and immunity, is expressed as a latent complex (L-TGF- $\beta$ ) non-covalently associated with its prodomain and presented on immune cell surfaces by covalent association with GARP. Binding to integrin  $\alpha$ v $\beta$ 8 activates L-TGF- $\beta$ 1/GARP. The dogma is that mature TGF- $\beta$  must physically dissociate from L-TGF- $\beta$ 1 for signaling to occur. Our previous studies discovered that  $\alpha$ v $\beta$ 8-mediated TGF- $\beta$  autocrine signaling can occur without TGF- $\beta$ 1 release from its latent form. Here, we show that mice engineered to express TGF- $\beta$ 1 that cannot release from L-TGF- $\beta$ 1 survive without early lethal tissue inflammation, unlike those with TGF- $\beta$ 1 deficiency. Combining cryogenic electron microscopy with cell-based assays, we reveal a dynamic allosteric mechanism of autocrine TGF- $\beta$ 1 signaling without release where  $\alpha$ v $\beta$ 8 binding redistributes the intrinsic flexibility of L-TGF- $\beta$ 1 to expose TGF- $\beta$ 1 to its receptors. Dynamic allostery explains the TGF- $\beta$ 3 latency/activation mechanism and why TGF- $\beta$ 3 functions distinctly from TGF- $\beta$ 1, suggesting that it broadly applies to other flexible cell surface receptor/ligand systems.

## INTRODUCTION

Transforming growth factor- $\beta$  (TGF- $\beta$ ) is a multifunctional cytokine with key roles in development, immunity, cancer, and fibrosis.<sup>1–3</sup> TGF- $\beta$  has three distinct gene products (TGF- $\beta$ 1, - $\beta$ 2, and - $\beta$ 3) all expressed in inactive (latent) forms (L-TGF- $\beta$ ), and “activation” is essential for function.<sup>4</sup> Most therapeutic TGF- $\beta$  targeting strategies have not focused on specific latency and/or activation mechanisms but rather on global inhibition of TGF- $\beta$  signaling and have significant toxicities.<sup>3</sup> Improved understanding of latency and activation may facilitate better therapeutic approaches targeting TGF- $\beta$ .

Latency of mature TGF- $\beta$  is determined by non-covalent association with its N-terminal prodomain cleaved by furin during

biosynthesis.<sup>5,6</sup> The prodomain encircles the mature TGF- $\beta$  homodimer in a ring-shaped disulfide-linked homodimer (latency-associated peptide [LAP]), forming L-TGF- $\beta$ .<sup>5</sup> LAPs serve four essential functions: (1) conferring latency through shielding mature TGF- $\beta$  from its receptors via the lasso loops of the strait-jacket domains<sup>5</sup>; (2) sequestering L-TGF- $\beta$  to the matrix or cell surface through binding to TGF- $\beta$  milieu molecules such as glycoprotein-A repetitions predominant (GARP), which stabilizes and covalently links L-TGF- $\beta$  to cell surfaces<sup>7–9</sup>; (3) facilitating proper folding and efficient secretion<sup>10</sup>; and (4) binding to essential activating proteins, in particular, integrins.<sup>1,11</sup>

Active mature TGF- $\beta$ s are disulfide-linked homodimers highly conserved in TGF- $\beta$  receptor (TGF- $\beta$ R) binding domains, particularly mature TGF- $\beta$ 1 and - $\beta$ 3, which bind with similar affinities to



TGF- $\beta$ Rs (TGF- $\beta$ R1/TGF- $\beta$ R2).<sup>12,13</sup> Despite this conservation, mice deficient in TGF- $\beta$ 1 or TGF- $\beta$ 3 have distinct phenotypes, potentially due to individual mechanisms of latency and/or activation as predicted by overall low homology between LAPs of TGF- $\beta$ 1 and - $\beta$ 3 (Figure S1).<sup>14–17</sup> Interestingly, both TGF- $\beta$ 1 and - $\beta$ 3 LAPs contain the integrin binding motif RGDLLXLI and bind to two integrins,  $\alpha$ v $\beta$ 6 and  $\alpha$ v $\beta$ 8, which together account for the majority of TGF- $\beta$ 1 and some of TGF- $\beta$ 3 function *in vivo*.<sup>11,18,19</sup> Integrin binding culminates in TGF- $\beta$  activation, leading to autocrine<sup>20</sup> or paracrine<sup>21</sup> TGF- $\beta$  signaling by mechanisms that remain speculative.<sup>18,19,22,23</sup>

Structural and sequence differences between integrins  $\alpha$ v $\beta$ 6 and  $\alpha$ v $\beta$ 8 suggest distinct mechanisms of TGF- $\beta$  activation likely contributing to context-specific functions of TGF- $\beta$ .<sup>11,18,19,22,24–26</sup> In the case of  $\alpha$ v $\beta$ 6, global conformational changes transduce force from the actin-cytoskeleton to L-TGF- $\beta$ , disrupting LAP and allowing the release of mature TGF- $\beta$  for paracrine signaling.<sup>23</sup> This mechanism requires the highly conserved  $\beta$ 6-subunit cytoplasmic domain, which binds the actin cytoskeleton.<sup>19</sup> However, released mature TGF- $\beta$ 1 from  $\alpha$ v $\beta$ 8-mediated activation is difficult to detect, indicating inefficient paracrine TGF- $\beta$ 1 signaling.<sup>22,27</sup> Accordingly,  $\alpha$ v $\beta$ 8 does not undergo global conformational changes,<sup>28,29</sup>  $\alpha$ v $\beta$ 8-mediated TGF- $\beta$ 1 activation does not require actin-cytoskeleton force generation since the  $\beta$ 8 cytoplasmic domain is not required for activation and does not bind to actin.<sup>18</sup> Our previous work revealed that  $\alpha$ v $\beta$ 8 binding induces flexibility in the L-TGF- $\beta$ 1 straitjacket, leading us to hypothesize that mature TGF- $\beta$ 1 can be activated without release from the latent complex, which we confirmed in cell-based assays.<sup>22</sup> Thus, we hypothesized that flexibility generated by binding L-TGF- $\beta$ 1 to  $\alpha$ v $\beta$ 8 is sufficient to expose mature TGF- $\beta$ 1 to TGF- $\beta$ Rs for autocrine signaling without being released.<sup>22</sup> Yet, it remains unclear how without mechanical force,  $\alpha$ v $\beta$ 8 binding mechanistically induces L-TGF- $\beta$  flexibility when L-TGF- $\beta$  is stabilized by binding to GARP and whether such a mechanism is physiologically relevant, as it is widely assumed that release and paracrine signaling of TGF- $\beta$  are required for its function.<sup>1</sup>

In this study, we first validate autocrine signaling without TGF- $\beta$ 1 release is physiologically relevant. We engineer knockin mice globally expressing only *tgfb1* with a mutated furin cleavage site that cannot release TGF- $\beta$ 1. TGF- $\beta$  signaling in these mice remains intact as they survive, breed, and are spared from lethal early tissue inflammation of TGF- $\beta$ 1 deficiency, proving that mature TGF- $\beta$  bound to its latent complex can be activated, bind to its receptors, and signal.<sup>30</sup> We next pursue the mechanism allowing TGF- $\beta$ 1 to bind to TGF- $\beta$ Rs without release. We describe a dynamic allosteric model whereby, upon binding to  $\alpha$ v $\beta$ 8, reduction of local conformational entropy around the L-TGF- $\beta$  RGD binding region increases conformational entropy around distal regions of L-TGF- $\beta$ /GARP, exposing mature TGF- $\beta$  to TGF- $\beta$ Rs without release. In support of this model, we determine structures of L-TGF- $\beta$ 3/GARP showing the degree of basal conformational entropy of L-TGF- $\beta$ 1 and - $\beta$ 3 not only determines the basal level of integrin-independent TGF- $\beta$  activation but also entropy available to drive integrin-dependent TGF- $\beta$  activation. Higher levels of integrin-mediated entropy change in L-TGF- $\beta$ 3 than - $\beta$ 1 result in paracrine release of mature TGF- $\beta$ 3

but not - $\beta$ 1, indicating isoform-specific mechanisms of autocrine and paracrine TGF- $\beta$  signaling. Furthermore, the direction of entropy redistribution can be manipulated by stabilizing different flexible domains of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP. Overall, our structural and cell-based approaches reveal a protein dynamic-based allosteric mechanism of redistributing conformational entropy at large distances across protein complexes that is actin cytoskeletal force-independent and determines autocrine and paracrine TGF- $\beta$  functions. Together, these results advance mechanistic understanding of latency and activation of TGF- $\beta$  family members, providing a roadmap for structural understanding of protein dynamic-mediated signal propagation through flexible cell surface proteins.

## RESULTS

### Autocrine TGF- $\beta$ 1 signaling without release prevents lethal tissue inflammation caused by global TGF- $\beta$ 1 deficiency

TGF- $\beta$  signals through both autocrine and paracrine mechanisms (Figure 1A). TGF- $\beta$ 1 deficient mice lack both autocrine and paracrine TGF- $\beta$ 1 signaling from all cells and die early of widespread tissue inflammation (Figure 1B).<sup>30</sup> This is attributed to TGF- $\beta$  signaling in T cells since this same phenotype is observed when TGF- $\beta$ Rs are deleted from T cells.<sup>31,32</sup> Whether T cells receive primarily autocrine or paracrine TGF- $\beta$ 1 signals is not well understood. Our recent structural and cell-based studies demonstrated that TGF- $\beta$ 1 release was not required for autocrine TGF- $\beta$ 1 signaling.<sup>22</sup> We test the physiological significance of this finding by creating mice with a mutation in the canonical furin recognition sequence (<sup>275</sup>RXRR<sup>278</sup>↓) in TGF- $\beta$ 1 (*tgfb1*<sup>R278A/R278A</sup>) that cannot cleave mature TGF- $\beta$ 1 from LAP and are thus only capable of autocrine but not paracrine signaling (Figures 1C–1F and S2). We hypothesize that if non-released mature TGF- $\beta$ 1 productively binds to TGF- $\beta$ Rs and induces autocrine signaling, mutant mice are rescued from universal early lethal tissue inflammation of TGF- $\beta$ 1 deficiency.<sup>30</sup> *Tgfb1*<sup>−/−</sup> mice begin to show signs of wasting by 10–14 days and die within 24 days (Figures 1G and 1H). *Tgfb1*<sup>R278A/R278A</sup> mice are phenotypically indistinguishable from *tgfb1*<sup>R278A/WT</sup> and wild-type (WT) littermates up to 240 days (at the time of this manuscript submission), showing similar post-natal survival, and weight gain (Figures 1H–1J). Genetic approaches to determine the *in vivo* role of TGF- $\beta$ 1 are confounded by contributions of maternal endocrine TGF- $\beta$ 1 supplied transplacentally during development and after birth through breast milk. Maternal-derived TGF- $\beta$ 1 from *tgfb1*<sup>−/−</sup> dams partially compensates for fetal TGF- $\beta$ 1 deficiency, allowing *tgfb1*<sup>−/−</sup> mice to be born alive and survive until weaning before succumbing to autoimmunity.<sup>30,33</sup> When maternal TGF- $\beta$ 1 is absent, *tgfb1*<sup>−/−</sup> mice die immediately after birth.<sup>34</sup> We demonstrate endocrine release of cleaved mature TGF- $\beta$  from maternal sources is dispensable since *tgfb1*<sup>R278A/R278A</sup> mice can be derived from homozygous *tgfb1*<sup>R278A/R278A</sup> dams and show similar post-natal survival, gain weight, and are phenotypically indistinguishable from *tgfb1*<sup>R278A/R278A</sup> mice born from *tgfb1*<sup>R278A/WT</sup> dams (Figure 1J; Video S1). The organs of *tgfb1*<sup>R278A/R278A</sup> mice are histologically indistinguishable from WT and *tgfb1*<sup>R278A/WT</sup> mice, in contrast





with *tgfb1*<sup>-/-</sup> mice, which display massive immune infiltration of the heart, liver, and lungs (Figures 1K and 1L; Table S1). Therefore, autocrine TGF- $\beta$ 1 signaling without release rescues the early lethal tissue inflammation of TGF- $\beta$ 1 deficiency, and endocrine or paracrine release of TGF- $\beta$ 1 is not involved or required for this rescue.

To validate these findings, we performed several controls. We verified *tgfb1*<sup>R278A/R278A</sup> mice show no evidence of mature TGF- $\beta$ 1 cleavage (Figures 1M and 1N) and found non-cleaved TGF- $\beta$ 1 prominently expressed in WT lysates from organs and CD4+ T cells (Figures 1M and 1N) and easily detected on surfaces of WT CD4+ T cells, suggesting autocrine TGF- $\beta$ 1 signaling without release can also occur in WT mice (Figure 1O). We confirmed non-cleaved TGF- $\beta$ 1 induces sufficient TGF- $\beta$  signaling (Figure 1P) to generate immunosuppressive regulatory T cells (Tregs) *in vitro* (Figures 1Q–1S) and *in vivo* (Figures 1T–1V).

Taken together, our findings support the physiological relevance of autocrine TGF- $\beta$ 1 signaling without release of mature TGF- $\beta$ .

### Structures of the L-TGF- $\beta$ 1/GARP and $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP complexes

To address mechanisms allowing TGF- $\beta$ 1 to bind to TGF- $\beta$ Rs without release, we use single-particle cryogenic electron microscopy (cryo-EM) to study complexes of L-TGF- $\beta$ 1/GARP (Figures 2A and 2B) and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP in solution. By mixing L-TGF- $\beta$ 1/GARP with recombinant  $\alpha$ v $\beta$ 8 ectodomain

(1:1 molar ratio), we obtained anticipated proportions of 1:1 and 2:1  $\alpha$ v $\beta$ 8:L-TGF- $\beta$ 1/GARP complexes as revealed by mass photometry (Figure S3A) and single-particle cryo-EM (Figures 2C–2G and S3). Using a cell-based TGF- $\beta$ 1 activation assay, we demonstrated one  $\alpha$ v $\beta$ 8 is sufficient to activate TGF- $\beta$ 1 from L-TGF- $\beta$ 1/GARP for signaling (Figure S3B). Thus, while we determined structures for both 2:1 (Figure S3E) and 1:1  $\alpha$ v $\beta$ 8:L-TGF- $\beta$ 1/GARP complexes, we focused on the 1:1 complex obtaining a structure at 2.5 Å resolution (Figures 2C and S3E). By further intensive classification, we isolate a small percentage of unbound L-TGF- $\beta$ 1/GARP (4.6% particles, at 3.4 Å, Figure 2D) and  $\alpha$ v $\beta$ 8 (1.5% of particles, at 4.5 Å, Figure 2E), with remaining particles of the trimeric complex in many different conformations (93.9% of total particles, resolution 2.5–8.3 Å, Figures 2F–2H). In addition, we determined a 3.0 Å resolution structure of L-TGF- $\beta$ 1/GARP (Figure S3H) from the purified L-TGF- $\beta$ 1/GARP sample.

Overall, the cryo-EM structure of L-TGF- $\beta$ 1/GARP determined alone is largely consistent with its crystal structure (PDB: 6GFF),<sup>5</sup> except that we connect the straitjacket domain to the contralateral arm domain on the opposite side of L-TGF- $\beta$ 1. In most of the structures, the resolution is sufficient to resolve sidechains for reliable atomic model building (Figure 3A, left). The local resolution of the density map and the temperature factor (B-factor) of individual residues obtained from real-space refinement of the model are consistent (Figures 3B and S3E). We further subject L-TGF- $\beta$ 1/GARP to 1  $\mu$ s of all-atom molecular dynamics

(D) Schematic showing location of recombined targeted *tgfb1* locus on chromosome 7 (GenBank: M13177.1).

(E) Sequencing chromatograms of WT/WT, WT/KI, and KI/KI mice with corresponding sequence shown above.

(F) Schematic of one TGF- $\beta$ 1 protein monomer and position of LAP, furin cleavage site, mutated R278A sequence, and mature TGF- $\beta$ 1 peptide.

(G) Body weights of KO/KO mice at post-natal day 18 compared with littermate WT/KO controls. Shown is the mean and standard error (SEM). \* $p < 0.05$  Student's *t* test.

(H) KI/KI, WT/KI, and WT/WT mice survive to adulthood compared with KO/KO mice, which die by 24 days of multiorgan inflammation. \*\*\*\* $p < 0.0001$  of all groups compared with KO/KO by Mantel-Cox.

(I) Body weights over time of KI/KI, WT/KI, and WT/WT mice.

(J) Body weights over time of KI/KI mice born from KI/KI or WT/KI dams.

(K) Organ histology (heart, upper; lung, middle; and liver, lower) showing hematoxylin and eosin staining of representative fields of tissue sections from WT/WT (left), KI/KI (middle), and KO/KO (right) mice ( $n \geq 5$  mice). Scale bars, 30  $\mu$ m.

(L) Scatter plots of disease scores from mice KO/KO ( $n = 6$ ), WT/WT ( $n = 6$ ), WT/KI ( $n = 5$ ), and KI/KI ( $n = 6$ ) mice, represented by filled circles, open circles, open squares, or red filled triangles, respectively. Shown is mean, SEM. ANOVA followed by Tukey's multiple comparison test, \*\*\*\* $p < 0.0001$ .

(M) Anti-mature TGF- $\beta$ 1 immunoblot using equal amounts of plasma, or organ lysates (kidney, liver, lung, or spleen), under reducing conditions, from WT/WT, KO/KO, or KI/KI mice. + or – below indicate respective genotypes. Positions of molecular weight markers (MWs) on left. Expected positions of ~50 and 12.5 kDa uncleaved and cleaved TGF- $\beta$ 1 bands on right. Below, immunoblot using anti-actin as a protein loading control.

(N) CD4+ T cells from WT/WT or KI/KI mice cultured on BSA or immobilized  $\alpha$ v $\beta$ 8tr, as indicated below image. Mature TGF- $\beta$ 1 detected by immunoblotting as in (M). Upper: shorter exposure. Middle: longer exposure to show mature TGF- $\beta$ 1 in WT CD4+ T cells. Lower panel represents same membrane stripped and reprobed with anti-actin. Shown is a representative experiment ( $n = 3$ ).

(O) Non-cleaved TGF- $\beta$ 1 is present on the surface of WT CD4+ T cells. Activated WT mouse CD4+ T cell surface biotinylation, capture on streptavidin agarose (SA), and detection as above. Below, membrane stripped and reprobed with anti-Na+/K+ ATPase, as a cell membrane marker. Shown is a representative experiment of 3 with similar results.

(P) Same lysates from activated CD4+ T cells from (N), demonstrating (upper) increased  $\alpha$ v $\beta$ 8-mediated TGF- $\beta$  signaling detected by anti-phospho-SMAD2/3 (pSMAD2/3). Note pSMAD2 migrates slightly slower than pSMAD3, total SMAD2/3 (middle), and actin (lower). Shown is a representative experiment ( $n = 3$ ).

(Q) Cartoon showing generation of induced-Treg (iTreg).

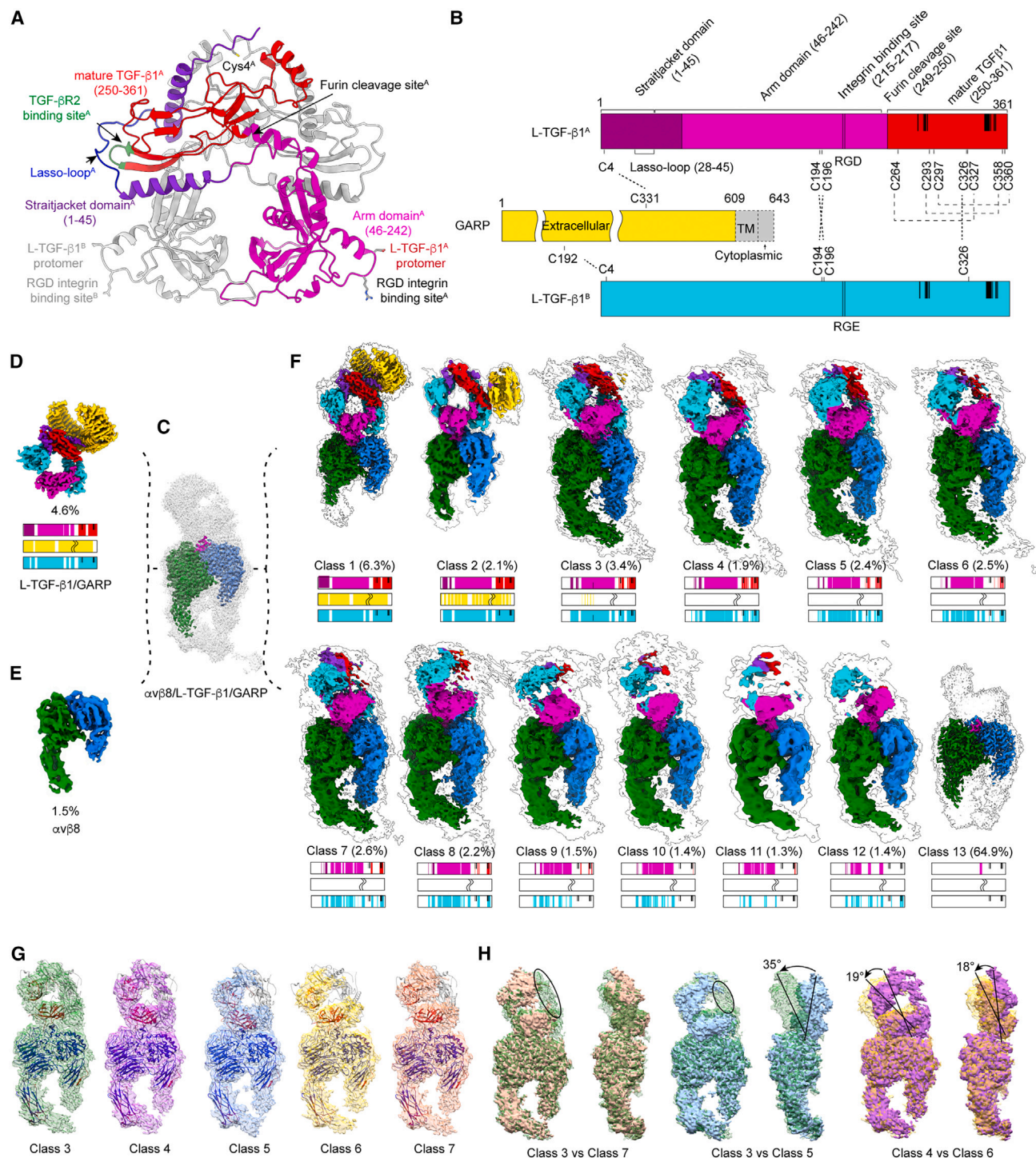
(R) CD4+ T cells in activating conditions from KI/KI compared with controls, WT/WT (or WT/KI) mice plated on BSA ( $\pm$  recombinant TGF- $\beta$ 1 as a positive control) or immobilized  $\alpha$ v $\beta$ 8 ectodomain, as indicated. CD4+ T cells are stained with anti-FoxP3 and anti-CD25, and representative quadrant scatterplots are shown. FoxP3+, CD25+ Treg in the upper right quadrant.

(S) Graphs showing Treg as percentage of activated CD4+ T cells. Shown is SEM. ns, not significant.

(T) Representative scatterplots of peripheral blood mononuclear cells (PBMCs) from KI/KI and age- and littermate-matched controls (WT/WT or WT/KI), as indicated, and stained as in (R).

(U and V) Treg enumerated from adult mouse PBMC ( $n = 9$  WT/WT and WT/KI;  $n = 8$  KI/KI) (U) or spleen at post-natal day ~18–21,  $n = 6$  WT/WT and WT/KI;  $n = 3$  KI/KI) (V) show no decrease in Treg compared with reduced Treg percentages in KO/KO mice (Figure S2). Shown is SEM. ns, not significant. \* $p < 0.05$ .

See also Figures S2 and S8.



**Figure 2. Structures and conformational flexibility of L-TGF-β1/GARP alone and bound with αvβ8**

(A) Atomic model of L-TGF-β1 dimer with domain in protomer A colored and marked.

(B) Schematic of L-TGF-β1/GARP constructs. Numbering starts after signal peptide. Two protomers and their respective RGD motif or RGE mutations, resolved cysteines, TGF-βR2 binding sites, and GARP transmembrane truncation indicated.

(C) Density maps of αvβ8/L-TGF-β1/GARP complex displayed with two thresholds, low in transparent gray, high in solid color. Color code: integrin αv-green, β8-blue.

(D and E) Density maps of L-TGF-β1/GARP (D) and αvβ8 (E) determined from classification of (C).

(legend continued on next page)

(MD) simulations, revealing a clear correlation between the per-residue root-mean-square-fluctuation (RMSF) and B-factor (Figure S4A). RMSF measures local structural flexibility and dynamics.<sup>35</sup> Thus, local resolution or B-factor provides quantitative measurement of relative flexibility of specific regions, which clearly shows half the straitjacket domain (including the lasso) is more flexible (Figure 3B, enlarged view in the upper) than the analogous portion of the other straitjacket within the same L-TGF- $\beta$ 1 (Figure 3B, enlarged view in lower). Our results suggest that, in solution, extensive interaction stabilizes the portion of the straitjacket domain in contact with GARP (Figure 3B, enlarged view in the upper), and exposure of mature TGF- $\beta$ 1 may require disruption of this extensive interaction.

### Structural dynamics and induced flexibility of $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP

Based on structures of L-TGF- $\beta$ 1/GARP alone and in complex with  $\alpha$ v $\beta$ 8, we hypothesized that integrin binding to L-TGF- $\beta$ 1/GARP further induces flexibility of GARP, L-TGF- $\beta$ 1, or both, leading to destabilization of the L-TGF- $\beta$ 1/GARP interface and lasso loops. In all snapshots that reflect motion and flexibility of GARP/L-TGF- $\beta$ 1 relative to  $\alpha$ v $\beta$ 8 (Figures 2F–2H and S3C–S3E), the domain close to the RGD binding loop is always resolved, but the density of the remaining part of L-TGF- $\beta$ 1/GARP is progressively weaker (Figure 2F). Despite only being resolved in two snapshots (Figure 2F classes 1 and 2), GARP is present in all L-TGF- $\beta$ 1 particles since a disulfide bond forms between GARP and each L-TGF- $\beta$ 1 monomer.<sup>9</sup> Extensive focused classification and alignment, together with 3D variability analysis (3DVA), reveal rocking motions of L-TGF- $\beta$ 1/GARP relative to  $\alpha$ v $\beta$ 8 (Figures 2G, 2H, S3F, and S3G; Video S2). Beyond rocking, we observe progressive loss of density from GARP to the straitjacket domain as range of motions increase (Figures 2G, 2H, and S4B). Visualizing both rocking and progressive changes of local resolution in GARP and the straitjacket rules out the possibility that loss of density is caused by particle misalignment rather than increased flexibility. Thus, we conclude that the disappearance of GARP in the reconstructions is caused by the increased flexibility of the straitjacket domain.

In one snapshot (class 1 in Figures 2F and 3A, right) where GARP is well resolved, which contains only 6.3% of classified particles, the lasso loop of the straitjacket domain interacting with GARP becomes more flexible after binding to  $\alpha$ v $\beta$ 8, as measured from both local resolution and change of normalized B-factor based on a common reference, while local resolutions of remaining portions of L-TGF- $\beta$ 1/GARP are comparable in structures of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP and L-TGF- $\beta$ 1/GARP (Figures 3B, 3C, and S3). As revealed in this best resolved structure of the trimeric complex, the arm domain of L-TGF- $\beta$  becomes more stable upon binding to  $\alpha$ v $\beta$ 8, indicated by a reduc-

tion of  $\sim 15 \text{ \AA}^2$  in B-factor from L-TGF- $\beta$ /GARP alone, but the straitjacket domain, including lasso loop and the interface of GARP with mature TGF- $\beta$ , becomes more flexible, with a  $\sim 15 \text{ \AA}^2$  increase in B-factor (Figure 3D). Consequentially, destabilization of the TGF- $\beta$ /GARP interface leads to progressive disappearance of L-TGF- $\beta$ /GARP in the reconstructions, also reflected as progressive increase of B-factor (Figures 2F classes 3 to 8, S4C, and S4D). Thus, binding to  $\alpha$ v $\beta$ 8 not only stabilizes the RGD loop and part of the arm domain that binds to the integrin but also allosterically induces more flexibility in distal regions of the L-TGF- $\beta$ 1 ring, particularly the lasso loop and straitjacket. These findings suggest that such induced flexibility activates TGF- $\beta$ 1 (Figures 3D and 3E).

### Spatial conformational entropy redistribution drives $\alpha$ v $\beta$ 8-mediated L-TGF- $\beta$ activation

What drives the allosteric activation of TGF- $\beta$ 1? The changes between L-TGF- $\beta$ 1/GARP and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP are not consistent with a “classic allostery” model conceptualized as a “domino effect” of conformational changes between stable structural endpoints.<sup>36</sup> In the best-resolved structures (Figure 3A, class 1), we observed minimal changes of L-TGF- $\beta$ 1 in its overall conformation after binding to integrin (GARP: root-mean-square deviation [RMSD] 1.3  $\text{\AA}$ , 3,950 atom pairs; L-TGF- $\beta$ 1 non-integrin binding subunit A: RMSD 1.9  $\text{\AA}$ , 2,496 atom pairs; and L-TGF- $\beta$ 1 integrin binding subunit B: RMSD 2.0  $\text{\AA}$ , 2,398 atom pairs). Rather, there are obvious changes in local resolution of reconstructed maps and per-residue B-factor in refined structures (Figures 3A–3D). Indeed, conformational flexibility instead of a series of discrete conformational changes is thought to drive dynamic allostery.<sup>37–40</sup>

Examining allostery through a thermodynamic lens allows connecting “classic” and “dynamic” allostery, where any change to the protein impacts free energy through both entropy and enthalpy. It is hypothesized that dynamic allostery influences free energy, predominantly via entropic contributions.<sup>37,41</sup> Based on the Boltzmann equation,  $S = k_B \ln(W)$ , where  $S$  is entropy,  $k_B$  is Boltzmann’s constant, and  $W$  represents the number of microstates,<sup>42</sup> higher conformational dynamics equal higher conformational entropy. It has also been observed that, upon binding small molecules, peptides, or DNA, proteins tend to redistribute their conformational entropy, i.e., reduce conformational entropy around the binding site and consequentially increase conformational entropy in a distal region.<sup>39,40,43,44</sup> As proposed previously,<sup>45,46</sup> spatial redistribution of conformational entropy explains dynamic allostery. Applying this concept to explain dynamic allosteric activation of L-TGF- $\beta$ 1, our results lead to a hypothesis whereby, upon binding to  $\alpha$ v $\beta$ 8, conformational entropy in L-TGF- $\beta$ /GARP is redistributed from the  $\alpha$ v $\beta$ 8 binding site to the L-TGF- $\beta$  straitjacket domain (Figure 3E),

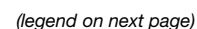
(F) Density maps of 13 sub-classes of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP, arranged from best to least resolved L-TGF- $\beta$ 1/GARP. Maps of classes 3–12 displayed at same two thresholds. Color scheme of L-TGF- $\beta$ 1 follows schematic in (A). Percentage below each map represents fraction of particles of the class. Bars are colored as in (A), with the unresolved regions shown in white.

(G) Five selected classes from (F) illustrating L-TGF- $\beta$ 1 motion relative to  $\alpha$ v $\beta$ 8. Ribbon diagram of  $\alpha$ v $\beta$ 8 and L-TGF- $\beta$ 1 docked within maps, arbitrarily colored.

(H) Comparison between sub-classes shown in (G) illustrate increased flexibility (class 3 vs. 7), degree, and direction of motion (class 3 vs. 5, and 4 vs. 6). Structures aligned to each other using  $\alpha$   $\beta$ -propeller domain showing L-TGF- $\beta$ /GARP rocking on top of  $\alpha$ v $\beta$ 8.

See also Figure S3.







allosterically exposing mature TGF- $\beta$  to TGF- $\beta$ R, leading to signaling.

We designed additional experiments to test this dynamic allosteric hypothesis. First, we tested whether stabilizing the L-TGF- $\beta$ /GARP interface in  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP would change the direction of spatial conformational entropy redistribution toward  $\alpha$ v $\beta$ 8. Using the inhibitory Fab MHG8, which binds to and stabilizes the L-TGF- $\beta$ /GARP interface,<sup>9</sup> we determine the  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP/MHG8 structure. Indeed, we find the L-TGF- $\beta$  straitjacket, including the LAP ring and integrin binding site is well resolved but most of  $\alpha$ v $\beta$ 8, including the head domain, is unresolved, confirming the redistribution of conformational entropy toward the integrin (Figures 3F–3H and S4E).

Following this experiment, we further tested if the direction of spatial conformational entropy redistribution can be altered by stabilizing flexible regions. We determined a cryo-EM reconstruction of L-TGF- $\beta$ /GARP in complex with full-length  $\alpha$ v $\beta$ 8 ( $\alpha$ v $\beta$ 8fl) reconstituted into lipid nanodisc ( $\alpha$ v $\beta$ 8fl-nd) constraining its otherwise flexible lower legs (Figures 3I, S4F, and S4G). Compared with class 1 of truncated  $\alpha$ v $\beta$ 8 ectodomain ( $\alpha$ v $\beta$ 8tr) bound with L-TGF- $\beta$ /GARP, this reconstruction has better resolved  $\alpha$ v $\beta$ 8 leg, but local resolution of L-TGF- $\beta$ /GARP is worse with higher B-factor (Figures 3J–3L and S4G). Together, our results suggest conformationally flexible regions in  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP serve as entropic reservoirs that can be regulated or manipulated to alter direction of entropy redistribution.

To further test this directionality of conformational entropy redistribution hypothesis, we constrained L-TGF- $\beta$ /GARP in

a cell membrane and allowed it to bind  $\alpha$ v $\beta$ 8 with various amounts of constraint, ranging from none to global stabilization (Figure 3M). In this system, L-TGF- $\beta$ /GARP is expressed in the cell membrane of a transformed mink lung epithelial TGF- $\beta$  responsive reporter cell (TMLC)<sup>47</sup> (Figure 3N) and allowed to bind with empty nanodisc as a control (Figure 3N, panel 1),  $\alpha$ v $\beta$ 8tr without constraint (2), C-terminally clasped  $\alpha$ v $\beta$ 8tr (3),  $\alpha$ v $\beta$ 8fl-nd (4), unclasped  $\alpha$ v $\beta$ 8tr globally stabilized by immobilization (5), or C-terminally clasped  $\alpha$ v $\beta$ 8tr globally stabilized by immobilization (6). For  $\alpha$ v $\beta$ 8tr, we predict that the membrane constraint imposed on L-TGF- $\beta$ /GARP directs entropy toward  $\alpha$ v $\beta$ 8, leading to inefficient TGF- $\beta$  activation (Figure 3M). Such constraints would be overcome by increasing the constraint imposed on  $\alpha$ v $\beta$ 8, leading to increasing the efficiency of TGF- $\beta$  activation (Figure 3M).

Indeed, with different forms of  $\alpha$ v $\beta$ 8 showing similar binding to L-TGF- $\beta$ /GARP (Figure S4H), soluble  $\alpha$ v $\beta$ 8tr without constraint does not efficiently induce TGF- $\beta$  signaling (Figure 3O). In comparison, soluble C-terminally clasped  $\alpha$ v $\beta$ 8tr or  $\alpha$ v $\beta$ 8fl-nd more efficiently activates TGF- $\beta$  signaling (Figure 3O). Global immobilized C-terminally clasped or unclasped  $\alpha$ v $\beta$ 8tr has the highest activation efficiency (Figure 3O). In this assay configuration, there is no mechanical force applied to  $\alpha$ v $\beta$ 8 from the actin cytoskeleton. Thus, the mechanism of  $\alpha$ v $\beta$ 8-dependent TGF- $\beta$  activation favors dynamic allostery. These experiments support the hypothesis that conformational entropy redistribution is not only sufficient but is the primary mechanism driving  $\alpha$ v $\beta$ 8-mediated L-TGF- $\beta$  activation.

### Figure 3. Spatial entropy redistribution upon L-TGF- $\beta$ /GARP binding to $\alpha$ v $\beta$ 8

- (A) Density maps of L-TGF- $\beta$ /GARP (left) independently determined and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP (right, class 1 from Figure 2F) docked with refined atomic model.
- (B and C) Ribbon diagram of L-TGF- $\beta$ /GARP (B) and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP (C). Residues colored by normalized B-factors with range indicated by scale bar. Two enlarged views show lasso loop (upper) and RGD containing arm domain (lower). Dashed loop, lower (B), indicates unresolved RGD loop.
- (D) Ribbon diagram of L-TGF- $\beta$  colored with normalized B-factor changes before and after binding to  $\alpha$ v $\beta$ 8, decreased (green), and increased (red). Arm domain binding to integrin becomes stabilized (lower dashed box with enlarged view), while straitjacket (containing the lasso, upper dashed box with enlarged view) and GARP become more flexible.
- (E) Model illustrating conformational entropy redistribution upon complex formation. Different regions are circled with colored dashed lines.  $\Delta S < 0$ , reduction of local entropy;  $\Delta S > 0$ , increase of local entropy; and  $\Delta S \approx 0$ , no change in local entropy. Blurring of ribbon diagrams indicates domain flexibility observed in structures.
- (F) Predicted spatial entropy redistribution upon L-TGF- $\beta$ /GARP/MHG8 binding to  $\alpha$ v $\beta$ 8. Labeling nomenclature same as (E).
- (G and H) Two different views of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP/MHG8 map (G) and docked with atomic model (H). L-TGF- $\beta$ /GARP/MHG8 is almost entirely resolved. Only a very small part of the integrin head domain is resolved.
- (I) Predicted spatial entropy redistribution upon L-TGF- $\beta$ /GARP binding to  $\alpha$ v $\beta$ 8fl-nd.
- (J) Comparison of local resolutions between reconstructions of  $\alpha$ v $\beta$ 8tr and  $\alpha$ v $\beta$ 8fl-nd in complex with L-TGF- $\beta$ /GARP. Local resolutions color coded by same scale. Both densities are displayed with two density thresholds.
- (K) Ribbon diagram of  $\alpha$ v $\beta$ 8fl-nd/L-TGF- $\beta$ /GARP. Residues colored by normalized B-factors with range indicated by scale bar. Enlarged views within dashed boxes show lasso loop (upper) and RGD containing arm domain (lower).
- (L) Ribbon diagram of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP colored with changes of normalized B-factor between  $\alpha$ v $\beta$ 8fl-nd and  $\alpha$ v $\beta$ 8tr in complex with L-TGF- $\beta$ /GARP, decreased (green) and increased (red).
- (M) Predicted entropy redistribution upon binding of soluble  $\alpha$ v $\beta$ 8tr,  $\alpha$ v $\beta$ 8fl-nd, and immobilized  $\alpha$ v $\beta$ 8tr to cell membrane-bound L-TGF- $\beta$ /GARP. Labeling nomenclature same as (E). Anchoring in cell membrane presumably increases stability of L-TGF- $\beta$ /GARP (1). Upon binding  $\alpha$ v $\beta$ 8tr, conformational entropy is redistributed from L-TGF- $\beta$  arm toward integrin  $\alpha$ v $\beta$ 8 (2). Stabilization of  $\alpha$ v $\beta$ 8tr via clasped and nanodisc reconstitution of  $\alpha$ v $\beta$ 8fl redistributes conformational entropy largely toward straitjacket domain (3 and 4). Immobilizing  $\alpha$ v $\beta$ 8tr drives entropy redistribution almost entirely toward the straitjacket, inducing sufficient flexibility for efficient TGF- $\beta$  activation (5 and 6).
- (N) Schematics showing design of TMLC reporter cell assays of TGF- $\beta$  activation without  $\alpha$ v $\beta$ 8 (1), soluble  $\alpha$ v $\beta$ 8tr without (2) or with C-terminal constraint (3), soluble  $\alpha$ v $\beta$ 8fl-nd (4), or globally stabilized immobilized  $\alpha$ v $\beta$ 8tr (5), or clasped  $\alpha$ v $\beta$ 8tr (6).
- (O) Activation of TGF- $\beta$  by soluble and immobilized  $\alpha$ v $\beta$ 8tr (clasped and unclasped),  $\alpha$ v $\beta$ 8fl-nd, and immobilized  $\alpha$ v $\beta$ 8tr using the assay configuration and numbering as in (N). Shown is the mean and standard error (SEM).

See also Figure S4.

### Intrinsic and induced flexibility of L-TGF- $\beta$ 3 and L-TGF- $\beta$ 3/GARP

L-TGF- $\beta$ 3 presented by GARP is essential during development and may also play a role in immunosuppressive immunity in post-natal life.<sup>48–51</sup> Therefore, we next study the structure and  $\alpha$ v $\beta$ 8-mediated activation of L-TGF- $\beta$ 3 alone and presented by GARP.

Using a similar strategy as for L-TGF- $\beta$ 1, we purified recombinant L-TGF- $\beta$ 3 and the L-TGF- $\beta$ 3/GARP complex (Figure 4A). Single-particle cryo-EM studies provided structures of L-TGF- $\beta$ 3/GARP (2.9 Å, Figures 4B and S5A), comparable with that of L-TGF- $\beta$ 1/GARP (3.0 Å). The portion of the straitjacket in contact with GARP almost identical in both structures (Figures 4C and S5B–S5D). However, L-TGF- $\beta$ 3 is significantly more flexible in all other regions by B-factor comparison to L-TGF- $\beta$ 1, particularly the arm, which contains the integrin binding site, and the portion of the straitjacket domain, including the lasso loop, that cradles the tip of mature TGF- $\beta$  containing the receptor binding domain (Figures 4D and 4E). Such increased intrinsic flexibility suggests that L-TGF- $\beta$ 3 is less constrained and contains higher basal entropy than L-TGF- $\beta$ 1. We hypothesize that increased basal entropy facilitates exposing mature TGF- $\beta$ 3 to TGF- $\beta$ Rs even without binding to  $\alpha$ v $\beta$ 8. After integrin binding, further entropic perturbation would lead to the release of mature TGF- $\beta$ 3 from its latent complex.

To test this hypothesis, we determined structures of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3 (2.7 Å) and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3/GARP (~4.9–7.2 Å) using L-TGF- $\beta$ 3 constructs where the furin cleavage site (R277A) was mutated to ensure mature TGF- $\beta$ 3 remained associated with the latent complex (Figures S5E and S5F). For image processing, we used the same procedure as applied to  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP, avoiding potential bias in data interpretation. Although  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3/GARP is stably formed (Figure 4F) and  $\alpha$ v $\beta$ 8 density is well resolved, only a small portion of L-TGF- $\beta$ 3 but no density of GARP is resolved (Figure 4G). To simplify structural analysis, we focused on  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3 without GARP (Figures 4H and S5F). Further classification reveals L-TGF- $\beta$ 3 rocks over the top of  $\alpha$ v $\beta$ 8 (Figure 4I) in a much larger range than L-TGF- $\beta$ 1 bound to  $\alpha$ v $\beta$ 8.<sup>22</sup> Indeed, in all conformational snapshots, the L-TGF- $\beta$ 3 straitjacket domains, including mature TGF- $\beta$ 3 peptides, are not resolved (Figures 4H and 4I). Overall, our structural studies of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3 and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3/GARP reveal similar but more dramatic redistribution of conformational entropy as seen in the  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP (Figure 2F class 1), since we could not isolate any subclass with either GARP or complete L-TGF- $\beta$ 3 (Figures S5E and S5F). Thus, we conclude that intrinsic flexibility of L-TGF- $\beta$ 3 is further enhanced upon  $\alpha$ v $\beta$ 8 binding by a similar conformational entropy redistribution mechanism as seen with L-TGF- $\beta$ 1/GARP (Figure 4J).

This presents a hypothesis that there is a threshold for flexibility of the straitjacket/lasso to allow mature TGF- $\beta$  to be exposed to its receptors without being released. If so, increased intrinsic flexibility of L-TGF- $\beta$ 3 presented by GARP allows mature TGF- $\beta$ 3 to be exposed to its receptors, allowing basal activation even without integrin binding. To test this, we expressed L-TGF- $\beta$ 3/GARP and measured TGF- $\beta$  activation using TMLC reporter cells, which, indeed, has significantly higher

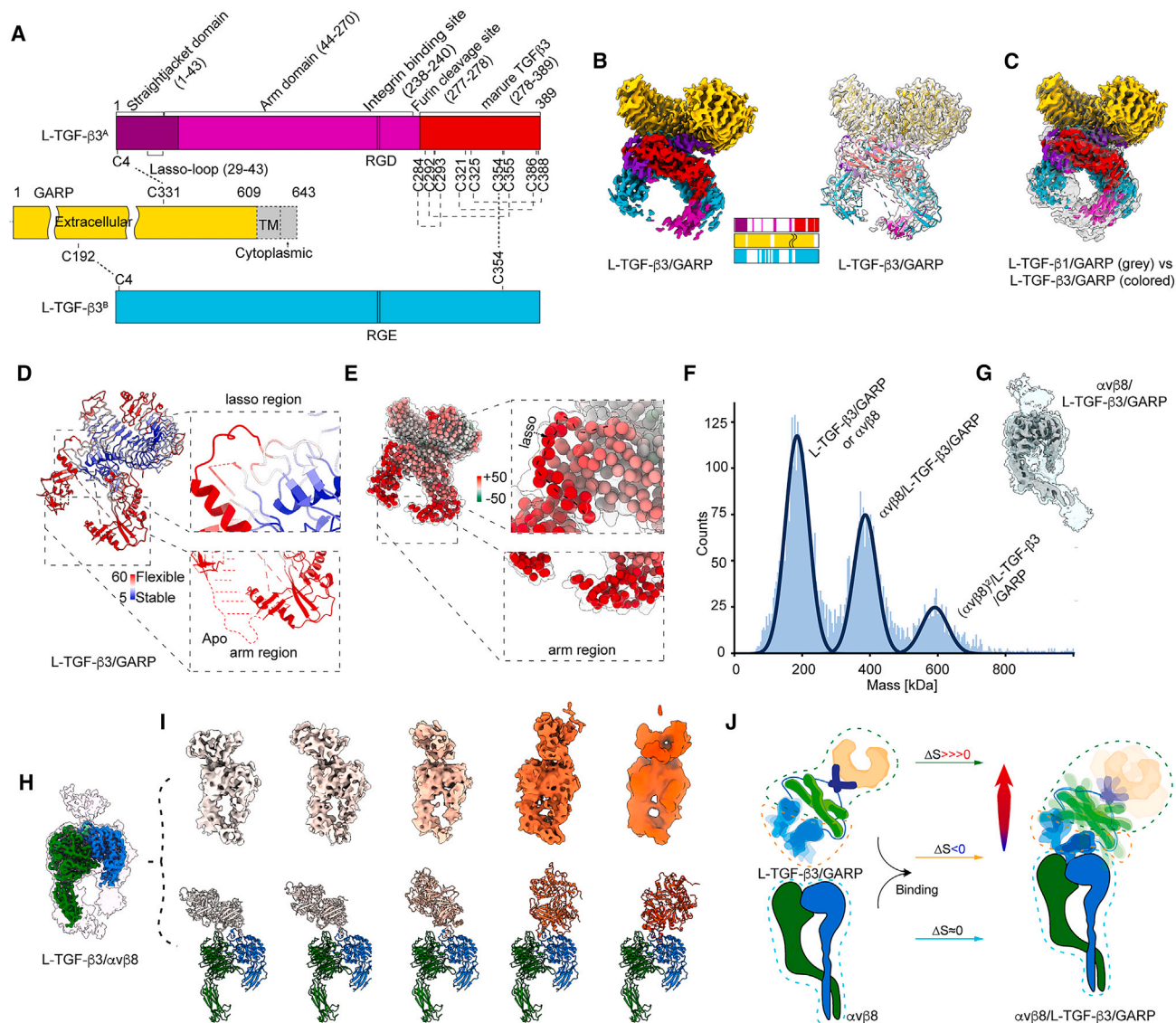
detectable basal TGF- $\beta$  activity than that of L-TGF- $\beta$ 1/GARP with no basal activity (Figures 5A, 5B, and S6). If there is similarly a threshold for flexibility of the straitjacket/lasso allowing mature TGF- $\beta$  to be released, higher induced flexibility of L-TGF- $\beta$ 3 upon  $\alpha$ v $\beta$ 8 binding could be sufficient to cause release of mature TGF- $\beta$ 3 (Figure 5C). Indeed, analysis of supernatant from L-TGF- $\beta$ 3/GARP TMLC cells cultured on immobilized  $\alpha$ v $\beta$ 8 contained significant amounts of released TGF- $\beta$  as opposed to supernatant from L-TGF- $\beta$ 1/GARP TMLC cells, which did not (Figures 5C and 5D).

Why mature TGF- $\beta$ 3 as opposed to TGF- $\beta$ 1 is released from its latent complex could be explained by relative differences in intrinsic flexibility of the lasso loop, a critical determinant of latency in all TGF- $\beta$  superfamily members.<sup>5,52</sup> A detailed comparison of sequence and structure reveals that the lasso loop of L-TGF- $\beta$ 3 (lasso3) is not only shorter than L-TGF- $\beta$ 1 (lasso1) but also less conserved in key residues interacting with mature TGF- $\beta$ <sup>53</sup> (Figure 5E). We thus hypothesize that lasso3 has evolved to be more flexible, providing less coverage to mature TGF- $\beta$ 3 from exposure to its receptor, allowing higher basal activity of L-TGF- $\beta$ 3. To test this hypothesis, we swapped the TGF- $\beta$ 3 lasso into TGF- $\beta$ 1 (L-TGF- $\beta$ 1\_lasso3) (Figure 5F) and observed significantly increased basal activation of TGF- $\beta$ 1 although not to the level of WT L-TGF- $\beta$ 3 (Figures 5G and S6). Together, we conclude that levels of conformational entropy of the arm, straitjacket, and lasso domains, as well as the structure of lasso loop, are key to maintaining latency, exposure, or release of mature TGF- $\beta$ .

### Functional consequences of TGF- $\beta$ release from $\alpha$ v $\beta$ 8-mediated L-TGF- $\beta$ activation

Under the physiological conditions where  $\alpha$ v $\beta$ 8-mediated TGF- $\beta$  activation occurs *in vivo*,  $\alpha$ v $\beta$ 8 is presented by one cell, but L-TGF- $\beta$  is presented on the cell surface of a contacting cell.<sup>27</sup> In this scenario,  $\alpha$ v $\beta$ 8-mediated TGF- $\beta$  activation could result in bidirectional signaling to both cells if TGF- $\beta$  was released (paracrine) or only in unidirectional signaling on the immune cell presenting TGF- $\beta$  cell if not released (autocrine). To test whether such  $\alpha$ v $\beta$ 8-mediated directional TGF- $\beta$  activation occurs, we devised an *in vitro* co-culture model system where the integrin  $\alpha$ v $\beta$ 8 is expressed by TGF- $\beta$ 1 null embryonic fibroblasts (MFB-F11) stably expressing a TGF- $\beta$  responsive secreted alkaline phosphatase (SEAP) reporter construct,<sup>54</sup> and TGF- $\beta$  is expressed on the surface of TMLC cells (Figures 6A–6C). The MFB-F11 reporter cells are highly sensitive to exogenous TGF- $\beta$ , indicating possession of the full complement of TGF- $\beta$ Rs and downstream signaling apparatuses.<sup>54</sup> When co-cultured with L-TGF- $\beta$ 1/GARP-expressing TMLC TGF- $\beta$  reporter cells, SEAP in cell supernatants reports TGF- $\beta$  signaling from  $\alpha$ v $\beta$ 8-expressing cells, while luciferase measured from cell lysates reports TGF- $\beta$  signaling from L-TGF- $\beta$ 1/GARP TMLC cells (Figures 6A–6C). Since MFB-F11 cells are TGF- $\beta$  deficient, the only cellular source of TGF- $\beta$ 1 in system is from L-TGF- $\beta$ 1/GARP-expressing TMLC TGF- $\beta$  reporter cells.

Co-culture of  $\alpha$ v $\beta$ 8-expressing MFB-F11 reporter cells with L-TGF- $\beta$ 1/GARP-expressing TMLC reporter cells results in autocrine signaling since only luciferase is detected (Figures 6A–6C). Such exclusivity of autocrine signaling can be attributed to



**Figure 4. Intrinsic flexibility of L-TGF- $\beta$ 3/GARP leads to high basal activation of TGF- $\beta$ 3**

(A) Schematic diagram of L-TGF- $\beta$ 3/GARP constructs, with all domains annotated and colored as in Figure 2A.

(B) Left: density map of L-TGF- $\beta$ 3/GARP with domains colored as in (A). Bars are colored following convention in (A), with the exception that unresolved regions are shown in white. Right: the same map (transparent) with ribbon diagram of L-TGF- $\beta$ 3/GARP displayed within.

(C) Comparison of maps of L-TGF- $\beta$ 1/GARP (transparent gray) and L-TGF- $\beta$ 3/GARP (colored solid surface) shows arm domain of L-TGF- $\beta$ 3/GARP is more flexible than L-TGF- $\beta$ 1/GARP.

(D) Ribbon diagram of L-TGF- $\beta$ 3 with residues colored by normalized B-factors with scale bar. Two enlarged views within dashed boxes show lasso loop (upper) and RGD containing arm domain (lower).

(E) A consensus model of L-TGF- $\beta$ 1/GARP and L-TGF- $\beta$ 3/GARP with each C $\alpha$  represented by a ball and colored with difference of normalized B-factors between L-TGF- $\beta$ 1/GARP and L-TGF- $\beta$ 3/GARP. Note B-factors of entire L-TGF- $\beta$ 3, particularly straitjacket (upper dashed box) and arm domains (lower dashed box), are much higher ( $\sim 50 \text{ \AA}^2$ ) than L-TGF- $\beta$ 1/GARP, indicating L-TGF- $\beta$ 3 is more flexible than L-TGF- $\beta$ 1 presented by GARP.

(F) Mass photometry histogram: peaks correspond to L-TGF- $\beta$ 3/GARP or  $\alpha$ v $\beta$ 8 alone ( $\sim 190 \text{ kDa}$ ), L-TGF- $\beta$ 3/GARP with one ( $\sim 390 \text{ kDa}$ ), or two  $\alpha$ v $\beta$ 8 integrins ( $\sim 590 \text{ kDa}$ ).

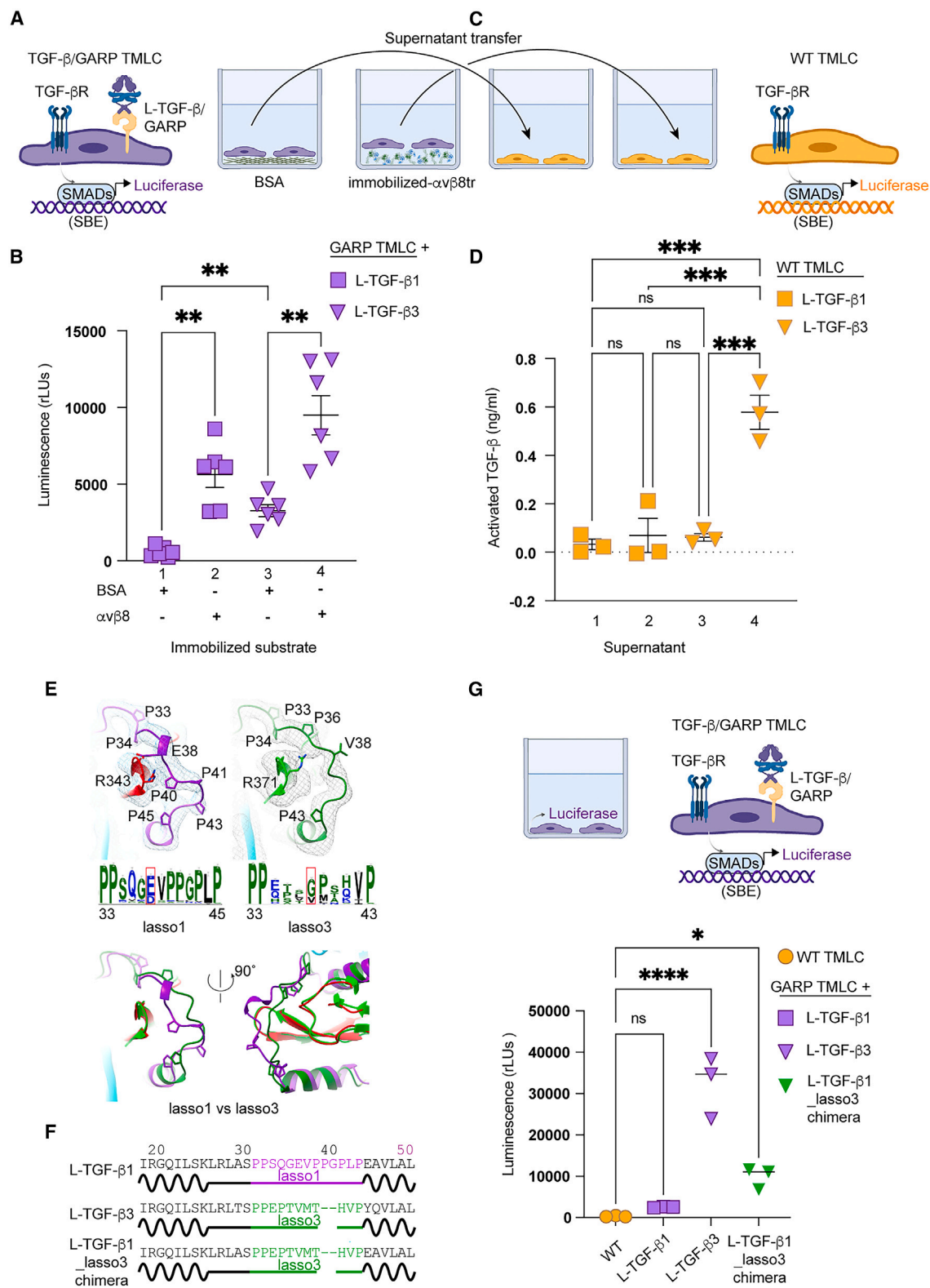
(G) Density map of L-TGF- $\beta$ 3/GARP bound with one  $\alpha$ v $\beta$ 8 at two thresholds. Disappearance of major part of L-TGF- $\beta$ 3/GARP indicates extensive flexibility upon binding to  $\alpha$ v $\beta$ 8.

(H) Density map reconstructed from all particles of L-TGF- $\beta$ 3 bound with one  $\alpha$ v $\beta$ 8. Map displayed at two thresholds.

(I) Upper row: 3D classification of all particles in (H) show flexibility of L-TGF- $\beta$ 3 bound with  $\alpha$ v $\beta$ 8. Bottom row: fitted atomic models of  $\alpha$ v $\beta$ 8 and L-TGF- $\beta$ 3 into corresponding maps shown in upper row.

(J) Cartoon of mechanistic model of intrinsic (left) vs.  $\alpha$ v $\beta$ 8-induced flexibility of L-TGF- $\beta$ 3/GARP (right). See also Figure S5.





**Figure 5.  $\alpha$ v $\beta$ 8 binding to L-TGF- $\beta$ 3 is sufficient to release mature TGF- $\beta$ 3**

(A) Cartoon of TGF- $\beta$  activation assay, with TMLC cells transfected and sorted to express equivalent levels of L-TGF- $\beta$ 1/GARP or L-TGF- $\beta$ 3/GARP on cell surfaces, cultured on either BSA or  $\alpha$ v $\beta$ 8 coated wells.

(legend continued on next page)

insufficient flexibility of the straitjacket and lasso loops of L-TGF- $\beta$ 1 for mature TGF- $\beta$ 1 to be released but sufficient to be exposed within the latent ring to bind to TGF- $\beta$ R2 after  $\alpha$ v $\beta$ 8 binding. The next question is whether directionality of L-TGF- $\beta$ 3 activation by  $\alpha$ v $\beta$ 8 is different than L-TGF- $\beta$ 1, since mature TGF- $\beta$ 3 is released upon  $\alpha$ v $\beta$ 8 binding (Figures 5C and 5D). Indeed, both autocrine and paracrine TGF- $\beta$ 3 signaling are observed since luciferase and SEAP are detected (Figure 6D). We hypothesize that within L-TGF- $\beta$ , mature TGF- $\beta$ 3 compared with TGF- $\beta$ 1 would be more accessible to TGF- $\beta$ R2, the first receptor binding mature TGF- $\beta$  to initiate signaling.<sup>55</sup>

To test whether TGF- $\beta$ R2 binds mature TGF- $\beta$  when exposed within L-TGF- $\beta$  complexes, we performed TGF- $\beta$ R2 binding assays to immobilized  $\alpha$ v $\beta$ 8-bound L-TGF- $\beta$ 3 or L-TGF- $\beta$ 1/GARP complexes. In these systems, the furin cleavage site between the mature TGF- $\beta$  and LAP is mutated at analogous positions as in *tgfb1*<sup>R278A/R278A</sup> mice, and thus mature L-TGF- $\beta$  cannot be released. Consistent with our structural analysis and TGF- $\beta$  activation assays, we observed more robust complex formation between TGF- $\beta$ R2 and L-TGF- $\beta$ 3/GARP than to L-TGF- $\beta$ 1/GARP when bound to immobilized  $\alpha$ v $\beta$ 8 (Figure 6E).

## DISCUSSION

### Physiological role of TGF- $\beta$ 1 activation without release

TGF- $\beta$ 1 plays major roles in mammalian biology from embryo implantation through the entire lifespan. For all roles, the dogma is that TGF- $\beta$  release is required for both autocrine and paracrine function.<sup>3</sup> This view is reinforced by numerous biochemical and structural experiments<sup>5,53,55–58</sup> but is challenged by our previous study of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1 predicting mature TGF- $\beta$ 1 can be sufficiently exposed to bind to its receptors within L-TGF- $\beta$ 1 without release.<sup>22</sup>

Here, we provide definitive evidence that mature TGF- $\beta$  without release supports autocrine signaling. In mice mature TGF- $\beta$ 1 covalently bound to LAP induces sufficient signaling to support immune function, since founders have so far survived 7 months without early immune lethality associated with global TGF- $\beta$ 1 deficiency.<sup>9,59–61</sup> Our ability to generate live births from homozygous *tgfb1*<sup>R278A/R278A</sup> intercrosses in the complete absence of WT maternal TGF- $\beta$ 1 from conception to adulthood provides definitive evidence that paracrine release of mature

TGF- $\beta$ 1 is not essential either for development or for early immune function.

Our results differ from previous reports where mice with furin conditionally deleted in T cells develop delayed organ inflammation similar to mice with conditional deletion of TGF- $\beta$ 1 in T cells.<sup>20</sup> However, furin potentially cleaves hundreds of substrates other than TGF- $\beta$ 1,<sup>62</sup> and likely has effects independent of TGF- $\beta$ 1. Mice with *tgfb1* conditionally deleted in T cells lack autocrine TGF- $\beta$ 1 signaling by T cells, whereas autocrine TGF- $\beta$ 1 signaling is preserved in *tgfb1*<sup>R278A/R278A</sup> mice. Confirmation that autocrine TGF- $\beta$ 1 signaling without release is sufficient to prevent autoimmunity not only validates our structure-based approach to studying the L-TGF- $\beta$  activation mechanism but demonstrates the power of cryo-EM to reveal structural mechanisms of flexible proteins that would otherwise have been unanticipated.

### L-TGF- $\beta$ activation driven by conformational entropy redistribution

The concept of conformational entropy redistribution, where entropy reduces around the ligand binding site and increases at distant sites is derived from conformational ensembles quantitatively characterized from structures obtained by X-ray crystallography or NMR spectroscopy of relatively small proteins.<sup>44,46</sup> With single-particle cryo-EM, protein conformational dynamics correlate with local resolutions of reconstructed structures, allowing exploration of conformational entropy redistribution in much larger and complex systems. Redistribution of conformational entropy also explains dynamic allosteric communication from ligand binding sites to distant sites without involving propagation of discrete stable conformational changes.<sup>38,46</sup>

Here, we used cryo-EM to examine conformational entropy redistribution in large and multi-component protein complex,  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ /GARP, where all components have been shown to be highly flexible in our previous<sup>22,28</sup> and current studies. By characterizing changes of conformational flexibility of different regions of L-TGF- $\beta$ /GARP induced by integrin  $\alpha$ v $\beta$ 8 binding, we show conformational entropy redistribution is the underlying dynamic allostery mechanism of  $\alpha$ v $\beta$ 8-mediated L-TGF- $\beta$  activation.

Specifically, we demonstrate that intrinsic flexibility, the basal conformational entropy of L-TGF- $\beta$  complexes, controls TGF- $\beta$  latency. In the case of the fully latent L-TGF- $\beta$ 1, the straitjacket and lasso loops are relatively stable (Figure 7A, panel 1), as

(B) TMLC cells expressing either L-TGF- $\beta$ 1/GARP (purple squares) or L-TGF- $\beta$ 3/GARP (purple inverted triangles) were cultured overnight on the indicated substrates, and luciferase activity was detected and reported as luminescence in relative light units (RLUs). Shown is the mean and standard error (SEM). \*\*\* $p$  < 0.01 by one-way ANOVA followed by Tukey's post-test.

(C) Cartoon of TGF- $\beta$  activation assay showing supernatants (A) (L-TGF- $\beta$ 1/GARP [yellow squares] or L-TGF- $\beta$ 3/GARP [yellow inverted triangles]) applied to wild-type (WT) TMLC cells detecting released TGF- $\beta$ .

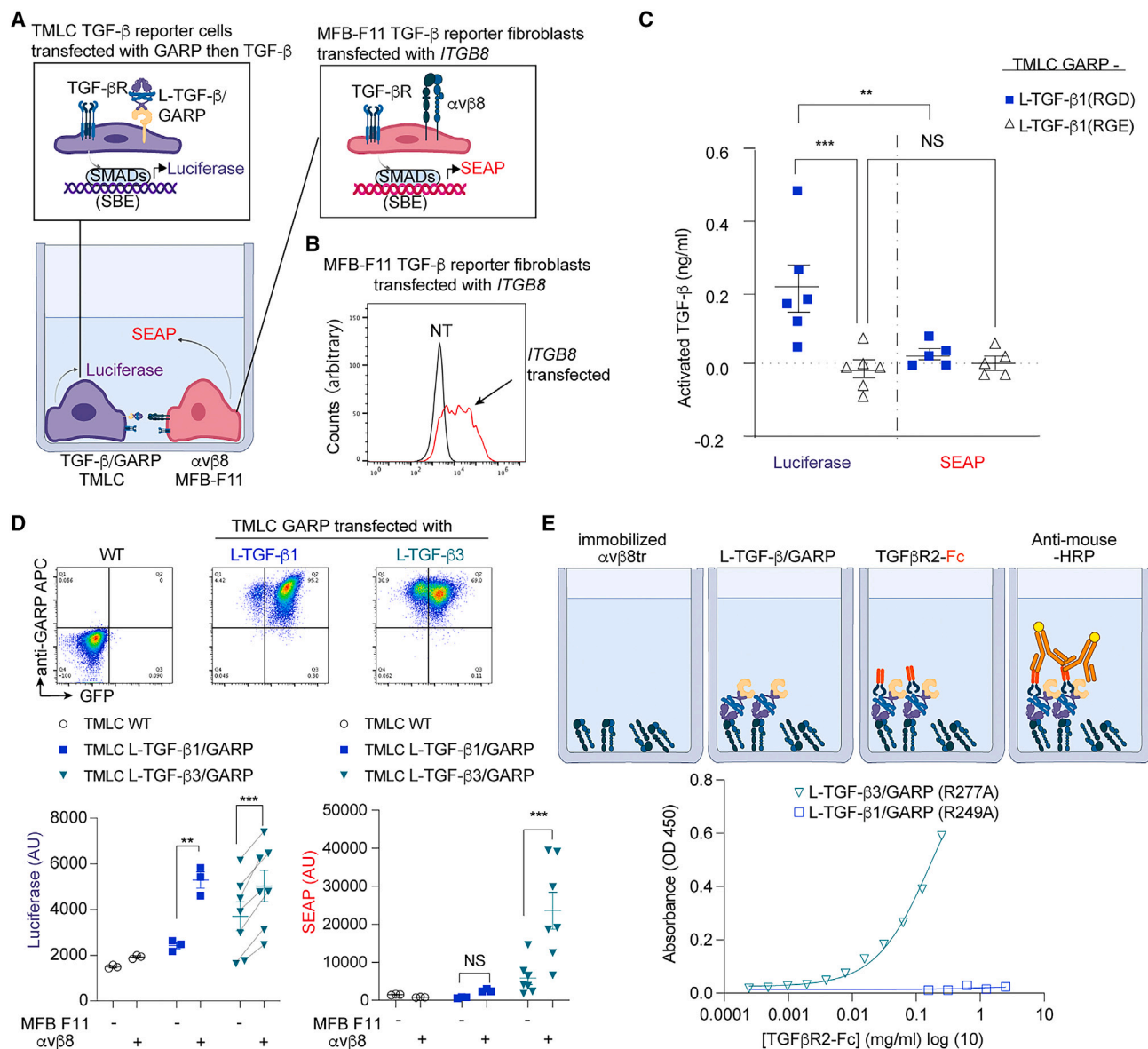
(D) Following overnight culture in format shown in (C) luciferase activity was detected. Results (C and D) shown as active TGF- $\beta$  (ng/mL). Shown is the mean and SEM. \*\*\* $p$  < 0.001 by one-way ANOVA followed by Tukey's post-test.

(E) Upper: ribbon models and filtered densities for L-TGF- $\beta$ 1 and L-TGF- $\beta$ 3 lasso-loops. Proline residues of lasso loops, used as landmarks, indicated. Middle: sequence position and species conservation (larger fonts indicate higher conservation) below ribbon models. Lower: overlays of L-TGF- $\beta$ 1 and L-TGF- $\beta$ 3 models in two views, illustrating lasso3 does not cover the TGF- $\beta$ R2 binding site of mature TGF- $\beta$  as effectively as lasso1.

(F) Sequence alignment showing lasso region of L-TGF- $\beta$ 1, - $\beta$ 3, and chimeric L-TGF- $\beta$ 1 with swapped lasso of L-TGF- $\beta$ 3.

(G) Lasso3 domain destabilizes L-TGF- $\beta$ 1/GARP. Upper: cartoon. Lower: TMLC stably expressing GARP transfected with constructs encoding L-TGF- $\beta$ 1 (purple square), L-TGF- $\beta$ 3 (purple inverted triangles), or L-TGF- $\beta$ 1 with swapped lasso3 (TGF- $\beta$ 1\_lasso3 chimera, green inverted triangles) and sorted for equivalent expression. WT TMLC- (yellow circles) or L-TGF- $\beta$ /GARP-expressing cell lines were cultured overnight, and luciferase activity was reported as luminescence (RLU). Shown is the mean and SEM. \* $p$  < 0.05, \*\*\*\* $p$  < 0.0001 by one-way ANOVA followed by Sidak's multiple comparison test.

See also Figure S6.



**Figure 6. Intrinsic and integrin-induced entropy of L-TGF- $\beta$  determines signaling directionality**

(A) Cartoon of design of dual TGF- $\beta$  reporter system, TMLC (purple) and MFB-F11 (red), is co-cultured (left). Shown are stably expressed reporter constructs with SMAD-binding elements (SBEs) driving indicated reporter proteins, luciferase, or secreted alkaline phosphatase (SEAP).

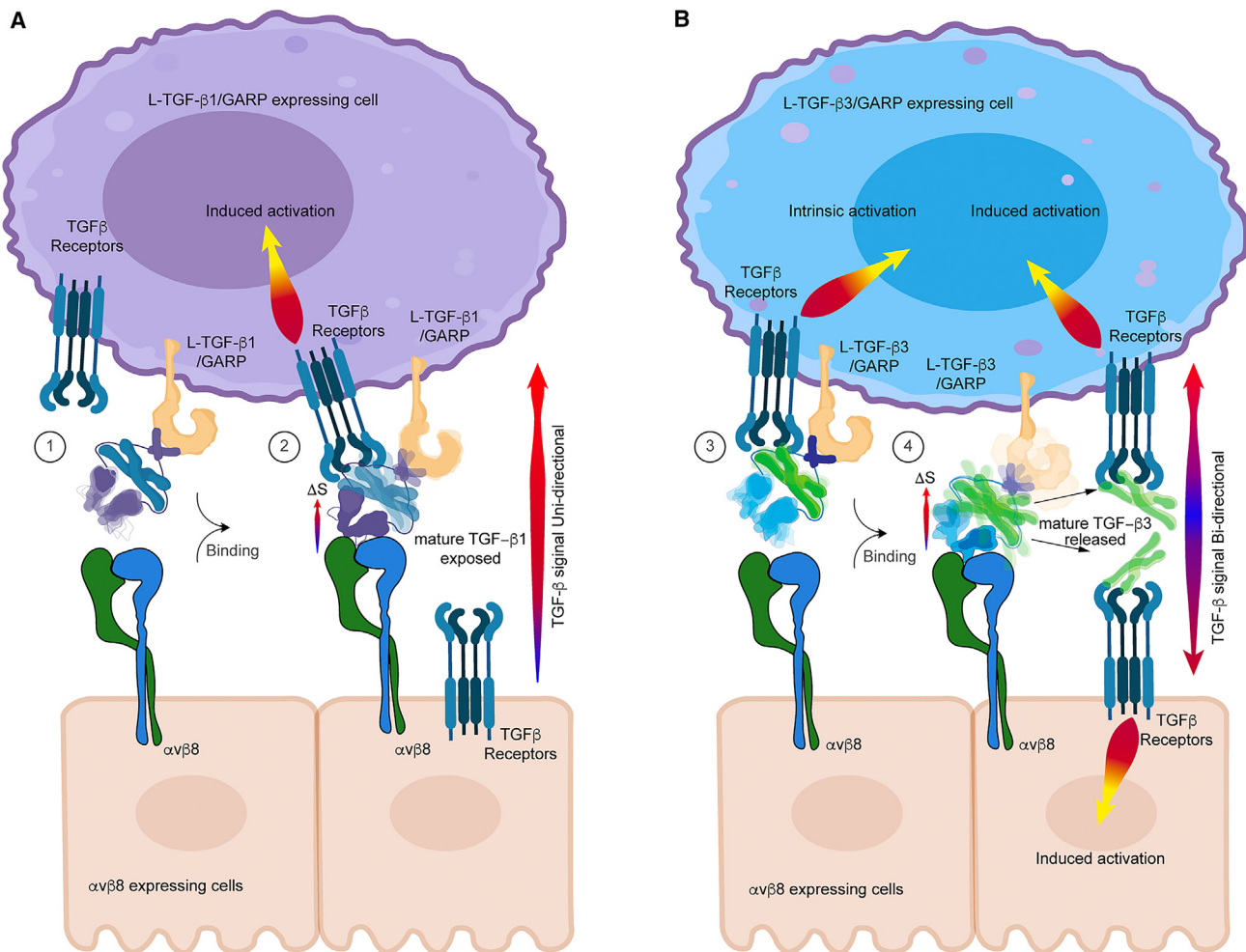
(B) MFB-F11 reporter cells stably transduced with an integrin  $\beta 8$  (*ITGB8*) expression construct sorted for high  $\alpha\text{v}\beta 8$  expression, using an anti- $\beta 8$  antibody. Histogram demonstrates expression of  $\alpha\text{v}\beta 8$  only seen in *ITGB8* transfected (red curve), not non-transfected (NT) cells (black curve).

(C) Mixing  $\alpha\text{v}\beta 8$ -expressing MFB-F11 cells with L-TGF- $\beta$ 1/GARP presenting TMLC cells is required and sufficient to activate TGF- $\beta$  signaling pathway on L-TGF- $\beta$ 1/GARP-expressing cells but not  $\alpha\text{v}\beta 8$ -expressing cells. L-TGF- $\beta$ 1(RGD)/GARP TMLC (filled squares), L-TGF- $\beta$ 1(RGE)/GARP TMLC (open triangles, characterized as Figure S3B). Results (vertical axis) normalized against standard TGF- $\beta$  activation curve. Shown is the mean and standard error (SEM). \*\* $p < 0.01$ , \*\*\* $p < 0.001$  by one-way ANOVA followed by Sidak's multiple comparison test.

(D) Upper: transfection of TMLC cells stably expressing GARP and co-transfected with either wild-type L-TGF- $\beta$ 1 or L-TGF- $\beta$ 3 IRES GFP plasmids. Surface expression of L-TGF- $\beta$ 1 and L-TGF- $\beta$ 3 are equivalent. Lower: co-culture of wild-type MFB-F11 (–),  $\alpha\text{v}\beta 8$  transfected (+) MFB-F11 cells (indicated below graph) with wild-type TMLC (black circles), TMLC expressing L-TGF- $\beta$ 1/GARP (blue filled squares), or L-TGF- $\beta$ 3/GARP (green triangles). Left: TMLC cells have significant basal levels of active TGF- $\beta$ 3 even when cultured without  $\alpha\text{v}\beta 8$ -expressing MFB-F11, but further increased by co-culture with  $\alpha\text{v}\beta 8$ -expressing MFB-F11 cells. Right: increased SEAP only seen when TMLC L-TGF- $\beta$ 3/GARP cells are co-cultured with  $\alpha\text{v}\beta 8$ -expressing MFB-F11. Results shown as mean and SEM in arbitrary units (a.u.). \*\* $p < 0.01$ , \*\*\* $p < 0.001$  by one-way ANOVA followed by Sidak's multiple comparison test, with the exception that results for L-TGF- $\beta$ 3/GARP cells on wild-type MFB-F11 or MFB-F11  $\alpha\text{v}\beta 8$ -expressing cells are shown as a paired t test.

(E) TGF- $\beta$ 2 binds more efficiently to  $\alpha\text{v}\beta 8$  bound L-TGF- $\beta$ 3/GARP compared with L-TGF- $\beta$ 1/GARP. Upper: cartoon showing sequential immobilization of  $\alpha\text{v}\beta 8$  ectodomain, binding of L-TGF- $\beta$ 1 or - $\beta$ 3/GARP complexes, TGF- $\beta$ 2-Fc (mouse-Fc), and anti-mouse-horseradish-peroxidase (HRP). Lower: representative experiment ( $n = 3$ ) shown with varying concentrations of TGF- $\beta$ 2-Fc, signal reported as optical density 450 nm (OD<sub>450</sub>).





**Figure 7. Dynamic entropy-based allosteric model of TGF-β activation**

(A) Cartoon of intrinsic and integrin-induced TGF-β1 activation. (1) L-TGF-β1/GARP has relatively low basal entropy in straitjacket/lasso, insufficient to trigger significant signaling. (2) Binding αvβ8 stabilizes arm domain but redistributes sufficient entropy to expose mature TGF-β1 to TGF-βRs to trigger signaling. (B) Cartoon of intrinsic and integrin-induced TGF-β3 activation. (3) TGF-β3/GARP has relatively high levels of basal entropy in straitjacket/lasso sufficient to expose mature TGF-β3 to TGF-βRs, allowing basal L-TGF-β3 constitutive autocrine signaling. (4) Binding to αvβ8 stabilizes the arm domain redistributing entropy exposing mature TGF-β3 to TGF-βRs, sufficient for paracrine release of mature TGF-β3 for bidirectional signaling to L-TGF-β3-presenting and αvβ8-expressing cells.

opposed to the partial latent L-TGF-β3, where the same domains are flexible (Figure 7B, 3). Binding to αvβ8 stabilizes the flexible RGD loop on the arm domain of both L-TGF-β1 and -β3 and reduces local conformational entropy. Spatial redistribution of this entropy toward the straitjacket enhances flexibility of the respective lasso loops. For L-TGF-β1, lower basal conformational entropy results in less entropy redistribution toward the lasso domain, insufficient to release mature TGF-β but sufficient to expose it to its receptor (Figure 7A, 2). Without being released, TGF-β is restricted to autocrine signaling. By contrast, because L-TGF-β3 has higher basal conformational entropy, αvβ8 binding results in more entropy redistribution toward the lasso domain, resulting in L-TGF-β3 passing a flexibility threshold sufficient for releasing mature TGF-β3 (Figure 7B, 4). Released TGF-β3 is capable of both autocrine and paracrine signaling to either αvβ8 or L-TGF-β3/GARP presenting cells.

### Physiological relevance of dynamic allostery in the activation of the TGF-β3/GARP complex

TGF-β3, as opposed to TGF-β1, is required for palatogenesis.<sup>13,16,63,64</sup> Our structural and cell-based findings of the αvβ8/L-TGF-β3/GARP complex are likely physiologically relevant since GARP and TGF-β3 form a covalent complex,<sup>9,50</sup> GARP and TGF-β3<sup>50</sup> colocalize to critical regions involved in palatogenesis, their respective genetic deficiencies in mice and/or humans lead to cleft palate,<sup>16,49,50,65</sup> and human genetic cleft palate syndromes result from missense mutations in the furin or the RGD sequence of *TGFB3*.<sup>14,66</sup> Interestingly, the cleft palate phenotype in *itgb8* null mice is only seen in a subset of live births.<sup>67,68</sup> We speculate that the intrinsic flexibility of L-TGF-β3 and exposure of mature TGF-β3 provides sufficient basal TGF-β signaling for palatogenesis even without integrin αvβ8 binding.

### Broader implication of the dynamic allostery mechanism in macromolecular complexes

Our data using the multicomponent  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 1(-\beta 3)/\text{GARP}$  model system provide evidence that redistribution of conformational entropy is a mode of allosteric regulation in a highly dynamic system. Protein dynamics are quantified as conformational entropy via the Boltzmann equation.<sup>42</sup> It has been demonstrated that protein dynamics can tune protein function without involving discrete conformational changes.<sup>38</sup> However, until very recently, examples of such dynamics were limited to side-chain rotamer ensembles by X-ray crystallography or methyl or amine group dynamics of small proteins by NMR spectroscopy.<sup>44</sup> The  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta/\text{GARP}$  provides a case study of a tunable functional endpoint (i.e., activation) correlating with protein dynamics in a relatively large complex. It is likely that dynamic allostery is a widespread mechanism to tightly regulate protein function, yet underappreciated, since methodology to decipher this new dimension in macromolecular protein function is only beginning to be applied. Single-particle cryo-EM is one valuable tool to directly visualize conformational dynamics in larger protein complexes. Combining it with other technologies, such as hydrogen-deuterium exchange mass spectrometry and/or molecular dynamics simulations, etc., it is reasonable to anticipate dynamic allostery driven by conformational entropy redistribution will be found to play important mechanistic roles in many biological systems.

### Intrinsic conformational entropy in the latency of the TGF- $\beta$ superfamily

The three TGF- $\beta$  isoforms are thought to be completely latent, along with a few others in the larger TGF- $\beta$  superfamily (i.e., growth differentiation factor-8 and -11 [GDF8 and GDF11]).<sup>5,52,69,70</sup> The high basal activity of L-TGF- $\beta 3$  was unexpected, which we attribute to the increased intrinsic entropy of straitjacket and lasso loops compared with TGF- $\beta 1$ . We propose that the relative intrinsic entropy of the straitjacket and lasso is a general evolutionary strategy controlling the degree of latency of TGF- $\beta$  superfamily members. Thus, as with L-TGF- $\beta 3$ , latency is clearly not absolute but rather determined by the degree of intrinsic conformational entropy. Extending this concept to non-latent TGF- $\beta$  superfamily members with available structures, all have highly flexible straitjacket and lasso loops, such as bone morphogenetic protein-9 (BMP9), BMP10, and activin A, suggesting they have very high levels of intrinsic entropy, allowing their growth factors to be freely exposed to receptors without requiring release (Figure S7). Overall, our data support an alternative hypothesis where extent of latency is a continuum controlled by levels of intrinsic entropy of the straitjacket, which when sufficiently high allows receptors to bind exposed receptor binding domains of mature growth factors while still within the prodomain complex. Among the TGF- $\beta$  superfamily, L-TGF- $\beta 1$ , with its relatively low entropy, appears to be an exception rather than the rule.

### Therapeutic implications

The general concept of latency of TGF- $\beta$  activation is binary, it is either latent or active. The binary dogmatic view has led to therapeutic approaches targeting released paracrine mature TGF- $\beta$

and have efficacy and safety issues.<sup>71–76</sup> The architecture and flexibility of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta/\text{GARP}$  suggest L-TGF- $\beta/\text{GARP}$  antibody binding epitopes are highly flexible and unstable, and multiple steric clashes limit access of TGF- $\beta$  traps or antibody inhibitors to TGF- $\beta$  or TGF- $\beta$ R within the complex. Indeed, we have observed poor inhibitory activity of antibody inhibitors to TGF- $\beta$ , TGF- $\beta$ Rs, L-TGF- $\beta 1/\text{GARP}$ , or TGF- $\beta$ R traps for  $\alpha\text{v}\beta 8$ -mediated activation of TGF- $\beta$ .<sup>27</sup> Thus, it is not surprising that immuno-oncology clinical trials using approaches that target paracrine-released TGF- $\beta$  have been disappointing due to lack of efficacy.<sup>77</sup> Our results predict antibodies targeting L-TGF- $\beta$  itself might also face similar efficacy challenges in clinical trials if the activation mechanism is  $\alpha\text{v}\beta 8$ -dependent.<sup>9,78,79</sup>

Mechanistic insights revealed from our study suggest why TGF- $\beta$  function can be highly context dependent, given dynamic allostery determines where and when TGF- $\beta$  is activated, whether it signals as an autocrine factor while remaining associated with the latent complex or is released, and ultimately whether it mediates paracrine signaling. Such a mechanism determines if TGF- $\beta$  primarily directs signaling to TGF- $\beta$ -presenting or integrin-expressing cells and cells in proximity or at a distance. Thus, targeting TGF- $\beta$  activation is highly complex, but such complexity offers opportunities for targeting context-dependent directional TGF- $\beta$  activation, which can be achieved at multiple levels either through targeting basal entropy, entropic redistribution, or release. Importantly, entropy redistribution can be manipulated to occur in different directions, as demonstrated by our findings (Figures 3F–3O). It remains to be determined how targeting entropically driven mechanisms will affect different pathologic scenarios. Overall, our results provide a structural framework for developing therapeutic approaches to inhibit context-specific functions of different TGF- $\beta$ s and argue against one-size-fits-all targeting strategies.

### Limitations of the study

The *tgfb1*<sup>R278A/R278A</sup> mouse model is early in establishment, and full immune characterization is ongoing. Thus, we cannot exclude delayed effects on tissue inflammation due to lack of paracrine release of mature TGF- $\beta 1$  as mice age further.

Despite providing a conceptual framework for understanding the mechanism of dynamic allostery, our current description of conformational entropy redistribution is only partially quantitative. More comprehensive quantitative descriptions of large-scale conformational entropy redistribution require major advancements of current methodologies.

### RESOURCE AVAILABILITY

#### Lead contact

Further information and requests for reagents may be addressed to Yifan Cheng (yifan.cheng@ucsf.edu).

#### Materials availability

All new materials generated in this manuscript, including mice, antibodies, cell lines, and plasmids, are available on request from the lead contact with a completed Materials Transfer Agreement.

#### Data and code availability

- All cryo-EM density maps, coordinates for the atomic models, and local-refined maps generated in this study have been deposited and are

publicly available. Accession numbers (EMDB, PDB IDs) are listed in the [key resources table](#).

- No new code was included in this study.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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## AUTHOR CONTRIBUTIONS

M.J., R.I.S., S.L.N., and Y.C. conceptualized the project. S.L.N. designed and oversaw the engineering of *tgfb1<sup>R278A</sup>/tgfb1<sup>R278A</sup>* mice. S.L.N., R.I.S., W.W., G.C., and T.S. designed, established, and performed mouse experiments. J.J., G.C., J.L.B., R.I.S., W.W., S.L.N., and N.D.C. performed and analyzed immunology experiments. M.J., S.L.N., and S.I. generated constructs. M.J., L.W., A.C., and J.L. purified protein samples. M.J. and A.C. prepared cryo-EM sample. M.J. conducted mass photometry. M.J., M.G.C., and Z.Y. performed data collection. M.J. performed cryo-EM data processing, model building, structure analysis, and TGF- $\beta$ R2 binding assay. R.I.S. made the cell lines. R.I.S., G.C., and T.S. performed TGF- $\beta$  activation assay. R.I.S. performed antibody binding assays. S.L.N., A.C., J.L., and J.M. designed, engineered, and produced clone F12. S.A.W. provided conceptual framework for protein dynamics. P.C. and P.K.T. performed MD simulations. M.J., R.I.S., S.L.N., and Y.C. wrote the manuscript with input from all authors.

## DECLARATION OF INTERESTS

S.L.N. is on the scientific advisory board (SAB) of Corbus Pharmaceuticals, LLC. Y.C. is on the SABs of ShuiMu BioSciences Ltd. and Pamplona Therapeutics. A provisional patent has been filed by the Regents of the University of California.

## STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- Assays to measure TGF- $\beta$ 1 activation by different forms of  $\alpha$ v $\beta$ 8
- Assays to measure TGF- $\beta$ 3 activation
- Assays to measure the signaling direction of the  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP complex
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- Cell surface biotinylation

## • QUANTIFICATION AND STATISTICAL ANALYSIS

## SUPPLEMENTAL INFORMATION

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## STAR★METHODS

### KEY RESOURCES TABLE

| REAGENT or RESOURCE                              | SOURCE                        | IDENTIFIER                         |
|--------------------------------------------------|-------------------------------|------------------------------------|
| <b>Antibodies</b>                                |                               |                                    |
| Anti-mouse TGF- $\beta$ 1                        | Abcam                         | ab179695                           |
| Anti-human, mouse $\alpha$ v $\beta$ 8, C6D4     | Takasaka et al. <sup>80</sup> | N/A                                |
| Anti-human $\alpha$ v, 8B8                       | Mu et al. <sup>18</sup>       | N/A                                |
| Anti-mouse HRP                                   | GE Healthcare                 | Cat. # NA931V; RRID:AB_772210      |
| Anti-LAP- $\beta$ 1-biotin or APC conjugated     | R&D Systems                   | Cat. # BAF246; RRID:AB_356332      |
| Anti-LAP- $\beta$ 1                              | R&D Systems                   | Cat. # AF426; RRID:AB_354419       |
| Anti-GARP                                        | BioLegend                     | Cat# 352502, RRID:AB_10897447      |
| Anti-human, mouse $\alpha$ v $\beta$ 8           | This paper                    | C6D4F12                            |
| anti-mouse-APC                                   | Biolegend                     | Cat# 405308, RRID:AB_315011        |
| anti-HA (clone 5E11D8)                           | Thermo Fisher                 | Cat# A01244, RRID:AB_1289306       |
| Anti-Na <sup>+</sup> /K <sup>+</sup> ATPase      | Thermo Fisher                 | Cat# MA5-32184, RRID:AB_2809472    |
| Anti-Actin                                       | Sigma                         | Cat# A2228, RRID:AB_476697         |
| Anti-pSMAD2/3                                    | Cell Signaling                | Cat# 8828, RRID:AB_2631089         |
| Anti-SMAD2/3                                     | Cell Signaling                | Cat# 3102, RRID:AB_10698742        |
| Live Dead Fixable Blue                           | Thermo                        | Cat#L23105                         |
| ACK lysis buffer                                 | Thermo                        | Cat#A1049201                       |
| Fc receptor block (CD16/32)                      | BD biosciences                | Cat# 553141, RRID:AB_394656        |
| Brilliant Staining Buffer                        | BD biosciences                | Cat# 563794                        |
| Foxp3 / Transcription Factor Staining Buffer Set | eBioscience                   | Cat#00-5523-00                     |
| Anti-mouse CD3                                   | Biolegend                     | Cat# 100201, RRID:AB_312658        |
| Anti-mouse CD45 (clone 30-F11)                   | Biolegend                     | Cat# 103128, RRID:AB_493715        |
| Anti-CD90.2 (clone 53-2.1)                       | BD biosciences                | Cat# 565257, RRID:AB_2739136       |
| Anti-mouse CD19 (clone 6D5)                      | Biolegend                     | Cat# 115566, RRID:AB_2832389       |
| Anti-mouse TCR $\beta$ (clone H57-597)           | Biolegend                     | Cat# 109249, RRID:AB_2810347       |
| Anti-mouse CD4 (clone RM4-5)                     | Biolegend                     | Cat# 100559, RRID:AB_2562608       |
| Anti-mouse CD8a (clone 53-6.7)                   | Biolegend                     | Cat# 100740, RRID:AB_2563055       |
| Anti-mouse CD25 (clone PC61)                     | Biolegend                     | Cat# 102072, RRID:AB_2894645       |
| Anti-mouse FoxP3 (clone MF-14)                   | Biolegend                     | Cat# 126405, RRID:AB_1089114       |
| CD4 <sup>+</sup> T cell isolation kit            | Miltenyi                      | Cat#130-104-454                    |
| Serum Free Medium, containing IMDM               | Gibco                         | Cat#12440-053                      |
| 1% Insulin-Transferrin-Selenium                  | Gibco                         | Cat#41400-045                      |
| 2-Mercaptoethanol                                | Gibco                         | Cat#31350-010                      |
| Retinoic acid                                    | Sigma                         | Cat#R2625                          |
| IL-2                                             | R&D Systems                   | Cat#402-ML/CF                      |
| Recombinant TGF- $\beta$ 1, human                | R&D Systems                   | Cat#240-B                          |
| Anti-mouse CD25-APC                              | Biolegend                     | Cat# 102012, RRID:AB_312861        |
| anti-mouse Foxp3-PE                              | Biolegend                     | Cat# 126404, RRID:AB_1089117       |
| RIPA buffer                                      | Sigma                         | Cat#R0278                          |
| Protease inhibitor cocktail                      | Thermo Scientific             | Cat#87786                          |
| Phosphatase inhibitor                            | Thermo Scientific             | Cat#A32957                         |
| BCA assay                                        | Thermo Scientific             | Cat#23228                          |
| Anti-Mouse-IgG                                   | Jackson ImmunoResearch        | Cat# 711-035-152, RRID:AB_10015282 |

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**Continued**

| REAGENT or RESOURCE                                          | SOURCE                       | IDENTIFIER                        |
|--------------------------------------------------------------|------------------------------|-----------------------------------|
| Anti-Rabbit-IgG                                              | Jackson ImmunoResearch       | Cat# 715-035-150, RRID:AB_2340770 |
| <b>Bacterial and virus strains</b>                           |                              |                                   |
| DH5a Chemically Competent E. coli                            | Thermo Fisher                | Cat. # 18265017                   |
| <b>Chemicals, peptides, and recombinant proteins</b>         |                              |                                   |
| Puromycin                                                    | Sigma Aldrich                | Cat. #P8833                       |
| Hygromycin                                                   | Thermo Fisher                | Cat. # 10687010                   |
| G418 sulfate                                                 | Thermo Fisher                | Cat. # 10131035                   |
| HRV-3C protease                                              | Millipore Sigma              | Cat. # 71493-3                    |
| Gibson Assembly Cloning Kit                                  | NEB                          | Cat. #E5510S                      |
| KAPA Mouse Genotyping Kit                                    | Sigma Aldrich                | KR0385_S – v3.20                  |
| Protein-G Agarose                                            | Pierce                       | Cat. # 20398                      |
| Strep-tactin agarose                                         | IBA                          | Cat. # 2-1204-001                 |
| Ni-NTA agarose                                               | Qiagen                       | Cat. # 30210                      |
| Lipofectamine 3000                                           | Thermo Fisher                | Cat. # L3000001                   |
| rhTGF- $\beta$ 1                                             | R&D systems                  | Cat. # #240-B-002/CF              |
| NHS-LC Biotin                                                | Thermo                       | Cat#A39257                        |
| <b>Critical commercial assays</b>                            |                              |                                   |
| Luciferase Assay System                                      | Promega                      | Cat. # E1500                      |
| SEAP Assay System                                            | Invitrogen                   | Cat. # T1017                      |
| <b>Experimental models: Cell lines</b>                       |                              |                                   |
| Mink: TMLC                                                   | Abe et al. <sup>47</sup>     | N/A                               |
| Mouse: MFB-F11                                               | Tesseur et al. <sup>54</sup> | N/A                               |
| Human: Expi293F                                              | Thermo Fisher                | Cat. # A14527                     |
| Hamster: ExpiCHO-S                                           | Thermo Fisher                | Cat. # A29127                     |
| Mouse: MFB-F11 with ITGB8                                    | This paper                   | N/A                               |
| Mouse: MFB-F11 with GARP and L-TGF- $\beta$ 1                | This paper                   | N/A                               |
| Mouse: MFB-F11 with GARP and L-TGF- $\beta$ 3                | This paper                   | N/A                               |
| Mouse: MFB-F11 with GARP and L-TGF- $\beta$ 1_Jasso3 chimera | This paper                   | N/A                               |
| Mouse: MFB-F11 with GARP and L-TGF- $\beta$ 1(RGE)           | This paper                   | N/A                               |
| Phoenix-AMPHO                                                | ATCC                         | CRL-3213                          |
| <b>Experimental models: Mice</b>                             |                              |                                   |
| Mouse: B6(Gt(ROSA)26Sortm1(PGK1-cre)Ozg                      | Ozgene                       | MGI:5435692                       |
| Mouse: 129X1/SvJ                                             | The Jackson Laboratory       | Strain #:000691                   |
| Mouse: C57BL/6J- <i>tgfb1</i> <sup>em2Lutzy/Mmjax</sup>      | The Jackson Laboratory       | Strain #:000691                   |
| Mouse: B6.129- <i>tgfb1</i> <sup>R278A/R278A</sup>           | This paper                   | N/A                               |
| <b>Oligonucleotides</b>                                      |                              |                                   |
| 5'- tgcacagtacctcatgcaca-3'                                  | This paper                   | JaxTGFB1F                         |
| 5'-gaacacagtgctaggcagg-3'                                    | This paper                   | mTGFB1Ex3R1                       |
| 5'-ctgtcctggaactcactctgtag-3'                                | This paper                   | TGFB1 1F                          |
| 5'-gtttggatgttggtgaagga-3'                                   | This paper                   | TGFB1 KI/cKI 4R                   |
| 5'-ccacatttgagaaggac-3'                                      | This paper                   | TGFB F WT only                    |
| 5'-catacattatcgaagtatgatctaag-3'                             | This paper                   | TGFB F KI only                    |
| 5'-gacatacacacacttagagg-3'                                   | This paper                   | TGFB WT/KI rev                    |
| 5'-gctcagttggctgttttgagg-3'                                  | This paper                   | ROSAWT F                          |

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**Continued**

| REAGENT or RESOURCE                                | SOURCE     | IDENTIFIER          |
|----------------------------------------------------|------------|---------------------|
| 5'-tagaacagctaaaggtagtgc-3'                        | This paper | ROSAFlp F           |
| 5'-atttcggcgctaaggatgactc-3'                       | This paper | ROSAcre F           |
| 5'-ttacacctgttcaattccctg-3'                        | This paper | ROSA R              |
| 5'-ctgaaccaaggagacggaatac-3'                       | This paper | tgfb1 Ex3/4 cDNA F  |
| 5'-gtgttagagggcaaggaccttg-3'                       | This paper | tgfb1 ex 6/7 cDNA R |
| 5'-gcaacaacgccatctatgag-3'                         | This paper | tgfb1 Ex1/2 cDNA F  |
| 5'-gctgatcccggtgatttcac-3'                         | This paper | tgfb1 Ex4/5 cDNA R  |
| 5'-cagggtgctgtaggctagcatcg-3'                      | This paper | hTGFB1 Stop F       |
| 5'-gcgccactagctcgagttatcag-3'                      | This paper | hTGFB1 Stop R       |
| 5'-ccctgagccaacgggtgatgaccacgtccccgaggccgtgctgc-3' | This paper | Lasso3 SOE F        |
| 5'-tgccgtagtagtcggcctc-3'                          | This paper | hTGFB1 Bsu36I R     |
| 5'-ccatttcagggtgctgtaggc-3'                        | This paper | hTGFB1 F            |
| 5'-gtcatcaccgttggtcagggggctggtgagccgcagcttgacag-3' | This paper | Lasso3 SOE R        |
| 5'-ctctgataccaagctggctagccacc-3'                   | This paper | SBPHIS F            |
| 5'-cagggcacttctgttggtgaggacctgaacagcacctc-3'       | This paper | SBPHIS SOE R        |
| 5'-ttagaggtgctgtttcagggctctcaccaagacaaagtgcctg-3'  | This paper | HAGARP SOE F        |
| 5'-ccgctgtacaggctgttccc-3'                         | This paper | HAGARP R            |
| 5'-agggccgtgtggacgtgg-3'                           | This paper | HAGARP Tr F         |
| 5'-tctcctcgagttatcagttgatgttcttcagcccccttc-3'      | This paper | HAGARP Tr R         |
| 5'-ggggactgaagaacatcaacatgctgtactaccatcaccatc-3'   | This paper | GARP Spy F          |
| 5'-ggcttaccttgaaggcccttagctaccactggatccagta-3'     | This paper | GARP Spy R          |
| 5'-ccatgtcacaccttcagccc-3'                         | This paper | TGFB3 R277A F       |
| 5'-gtccaaagccgcttctcctctg-3'                       | This paper | TGFB3R277ASOER      |
| 5'-cagaggaagaaggcggcttgagac-3'                     | This paper | TGFB3R277ASOEF      |
| 5'-gtgtgtacagctccagcacc-3'                         | This paper | TGFB3 R277A R       |
| 5'-ggagaactggggcgctcaag-3'                         | This paper | TGFB3 RGE SOE F     |
| 5'-ggcgcccagttctccacgg-3'                          | This paper | TGFB3 RGE SOE R     |
| 5'-gtgtgtacagctccagcacc-3'                         | This paper | TGFB3 R2            |
| 5'-ctctacgcgtactagtggcgcgccgg-3'                   | This paper | GFP F               |
| 5'-ttactgtacagctcgtccatgcc-3'                      | This paper | GFP R               |
| 5'-gactcactataggagaccaagctgg-3'                    | This paper | TGFB3 N Term F      |
| 5'-gtccaagtggtgcaagtggacagggaacctgaaac-3'          | This paper | TGFB3SOER           |
| 5'-ctgtccactgcaccaccttgac-3'                       | This paper | TGFB3SOEF           |
| 5'-ggtgagcctaagcttgctcaagatctg-3'                  | This paper | TGFB3 N Term R      |
| 5'-gtccaagtggtgctagtggacagggaacctgaaac-3'          | This paper | TGFB3C4SSOER        |
| 5'-ctgtccactagcaccaccttgac-3'                      | This paper | TGFB3C4SSOEF        |

**Recombinant DNA**

|                                                       |                                |                         |
|-------------------------------------------------------|--------------------------------|-------------------------|
| Human TGF- $\beta$ 1_pLX307                           | Rosenbluh et al. <sup>81</sup> | Addgene, Plasmid #98377 |
| Human TGF- $\beta$ 1 RGE_IRES2 EGFP puro pLX307       | This paper                     | N/A                     |
| Human TGF- $\beta$ 1 RGD_R249A_IRES2 EGFP puro pLX307 | This paper                     | N/A                     |
| Human TGF- $\beta$ 1 RGE_R249A_IRES2 EGFP puro pLX307 | This paper                     | N/A                     |
| Human L-TGF- $\beta$ 1_RGD_Lasso3 puro pLX307         | This paper                     | N/A                     |
| Integrin $\alpha$ v truncated, $\alpha$ vTr pcDM8     | Nishimura et al. <sup>25</sup> | N/A                     |
| Integrin $\beta$ 8 truncated, $\beta$ 8Tr pcDNA6      | Nishimura et al. <sup>82</sup> | N/A                     |
| $\beta$ 8 cDNA pBABE puro                             | Cambier et al. <sup>24</sup>   | N/A                     |

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**Continued**

| REAGENT or RESOURCE                                   | SOURCE                           | IDENTIFIER               |
|-------------------------------------------------------|----------------------------------|--------------------------|
| HA-GARP pcDNA3                                        | Cuende et al. <sup>78</sup>      | N/A                      |
| HIS SBP-GARP tr pcDNA6                                | This paper                       | N/A                      |
| HIS SBP-GARP tr SpyCatcher pcDNA6                     | This paper                       | N/A                      |
| Integrin $\alpha$ v full length, $\alpha$ vfl pCDVnRa | Nishimura et al. <sup>25</sup>   | N/A                      |
| Integrin $\beta$ 8 full length, pCD $\beta$ 8FNeo     | Nishimura et al. <sup>82</sup>   | N/A                      |
| SpyCatcher                                            | Keeble and Howarth <sup>83</sup> | Addgene, Plasmid #133447 |
| pLVE-hTGF $\beta$ 3-IRES-RED                          | Brunger et al. <sup>84</sup>     | Addgene, Plasmid #52580  |
| Human TGF- $\beta$ 3 RGD IRES RED pLX307              | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGD R277A IRES RED pLX307        | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGE R277A IRES RED pLX307        | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGE IRES RED pLX307              | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGD IRES GFP pLX307              | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGD R277A IRES GFP pLX307        | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGE R277A IRES GFP pLX307        | This paper                       | N/A                      |
| Human TGF- $\beta$ 3 RGE IRES GFP pLX307              | This paper                       | N/A                      |
| Human TGF- $\beta$ R2-Fc                              | Seed et al. <sup>27</sup>        | N/A                      |

**Software and algorithms**

|                |                                   |                                                                                                                                                                             |
|----------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MotionCor2     | Zheng et al. <sup>85</sup>        | <a href="https://msg.ucsf.edu/software">https://msg.ucsf.edu/software</a> ; RRID:SCR_016499                                                                                 |
| Relion 3.0     | Zivanov et al. <sup>86</sup>      | <a href="https://cam.ac.uk/relion">https://cam.ac.uk/relion</a> ; RRID:SCR_016274                                                                                           |
| SerialEM       | Mastronarde <sup>87</sup>         | <a href="http://bio3d.colorado.edu/SerialEM/">http://bio3d.colorado.edu/SerialEM/</a> ; RRID:SCR_017293                                                                     |
| cryoSPARC      | Punjani et al. <sup>88</sup>      | <a href="https://cryosparc.com">https://cryosparc.com</a> ; RRID:SCR_016501                                                                                                 |
| PyEM           | Daniel Asarnow, Yifan Cheng Lab   | <a href="https://github.com/asarnow/pyem">https://github.com/asarnow/pyem</a> ; <a href="https://doi.org/10.5281/zenodo.3576630">https://doi.org/10.5281/zenodo.3576630</a> |
| UCSF Chimera   | Pettersen et al. <sup>89</sup>    | <a href="https://www.cgl.ucsf.edu/chimera">https://www.cgl.ucsf.edu/chimera</a> ; RRID:SCR_004097                                                                           |
| UCSF ChimeraX  | Meng et al. <sup>90</sup>         | <a href="https://www.cgl.ucsf.edu/chimerax/">https://www.cgl.ucsf.edu/chimerax/</a> ; RRID:SCR_015872                                                                       |
| COOT           | Emsley et al. <sup>91</sup>       | <a href="https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/">https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/</a> ; RRID:SCR_014222                                 |
| PHENIX         | Adams et al. <sup>92</sup>        | <a href="http://www.phenixonline.org">http://www.phenixonline.org</a> ; RRID:SCR_014224                                                                                     |
| Clustal Omega  | Madeira et al. <sup>93</sup>      | <a href="https://www.ebi.ac.uk/Tools/msa/clustalo/">https://www.ebi.ac.uk/Tools/msa/clustalo/</a> ; RRID:SCR_001591                                                         |
| Prism 9        | GraphPad Software, San Diego, CA. | <a href="https://www.graphpad.com/scientific-software/prism/">https://www.graphpad.com/scientific-software/prism/</a> ; RRID:SCR_002798                                     |
| SpectroFlo     | CyTek Biosciences                 | <a href="https://cytekbio.com/pages/spectro-flo">https://cytekbio.com/pages/spectro-flo</a>                                                                                 |
| FlowJo™ v10.10 | Becton, Dickinson and Company     | <a href="https://www.flowjo.com">https://www.flowjo.com</a> RRID:SCR_008520                                                                                                 |

**Deposited data**

|                                            |            |                                                       |
|--------------------------------------------|------------|-------------------------------------------------------|
| $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP | This study | PDB: 8VSD, EMD-43494, EMD-43495, EMD-43496, EMD-43876 |
| L-TGF- $\beta$ 1/GARP                      | This study | PDB: 8VSC, EMD-43493                                  |
| L-TGF- $\beta$ 3/GARP                      | This study | PDB: 8VSB, EMD-43492                                  |
| $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3      | This study | PDB: 8VS6, EMD-43489                                  |

**Other**

|                                                                |            |                         |
|----------------------------------------------------------------|------------|-------------------------|
| QUANTIFOIL® R 1.2/1.3 on Au 300 mesh grids Holey Carbon Film   | Quantifoil | Product: N1-C14nAu30-01 |
| UltrAuFoil® R 1.2/1.3 on Au 300 mesh grids Holey Gold Supports | Quantifoil | N1-A14nAu30-01          |
| Quantifoil 400 mesh 1.2/1.3 holey carbon gold grid             | Ted Pella  | Q425AR-14               |
| Quantifoil 400 mesh 1.2/1.3 holey carbon copper grid           | Ted Pella  | 658-300-CU              |

## EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

### Mice

129SX1V/J x C57BL/6 *Tgfb1* fl/+ mice (Jax) with loxP3 sites flanking *tgfb1* exon 3 were crossed to 129X1SV/J x C57BL/6 *Rosa* 26-cre mice (Ozgene) to create *tgfb1*<sup>fl/+</sup> mice which were intercrossed to produce *tgfb1*<sup>-/-</sup> mice.<sup>33,94</sup> *Tgfb1* mice with a mutation in the furin cleavage site M13177.1c.1184-5AG>GC (p.Arg278Ala) were created at Ozgene (Perth, WA, Australia) using a conditional knock-in strategy on a C57BL/6 background (Figure S2). The targeting vector consisting of 5' homology arm containing *tgfb1* exon 3 followed a murine *tgfb1* cDNA minigene spanning exon 4-7, followed by a neomycin resistance cassette flanked by flippase recognition target sites (Frt), and the entire minigene and neo cassette flanked by loxP3 sites, which was inserted into intron 3, which was followed by exon 4, intron 4 and exon 5 with a mutation (AC to GC) in R278 to change the furin cleavage motif <sup>275</sup>RHRR<sup>278</sup> to <sup>275</sup>RHRA<sup>278</sup> (R278A) followed by a 3' homology arm. Successful targeting and germline transmission was followed by excision of the Frt flanked neo cassette to create a conditional KI (cKI) allele. Upon cre-mediated recombination, the loxP3 wild-type *tgfb1* exon 4-7 cDNA minigene can be excised and replaced with the *tgfb1* R278A mutant allele (Figure S2). C57BL/6 heterozygous (*tgfb1* cKI/+) or homozygous (*tgfb1* cKI/cKI) mice were crossed to C57BL/6 *Rosa* 26-cre mice (Ozgene). The resulting C57BL/6 KI/WT mice were mated to WT 129X1SV/J mice to generate 129X1SV/J x C57BL/6 *tgfb1*<sup>R278A/+</sup> mice. Alternatively, 129X1SV/J x C57BL/6 *tgfb1* cKI<sup>R278A</sup> /+ mice were crossed to 129X1SV/J x C57BL/6 *Rosa* 26-cre/WT mice (Ozgene) to create knock-in *tgfb1*<sup>R278A/+</sup> mice. Initial genotyping was performed using tail genomic DNA isolated and genotyped by PCR (Kapa) using primers TGFb1 1F, and TGFb1 KI/cKI 4R which produce a 654 bp band for the KI and 620 bp band for the WT allele. Subsequent genotyping was performed using WT or KI specific primers (TGFb F WT only or TGFb F KI only, paired with TGFb WT/KI rev) *Tgfb1*<sup>R278A/+</sup> mice breeding pairs from either strategy were intercrossed to produce *tgfb1*<sup>R278A</sup>/+*tgfb1*<sup>R278A</sup> mice. Mice were screened for flippase (Flp) *rosa* 26-cre using a primer mixture (ROSAWT F, ROSAFlp F, ROSAcre F, ROSA R) and flp + mice removed from the colony. WT, *tgfb1*<sup>R278A</sup>/*tgfb1*<sup>R278A</sup> or *tgfb1*<sup>R278A</sup>/*tgfb1*<sup>WT</sup> mice heterozygous or null for *Rosa* 26-cre and null for Flp were intercrossed and used for survival experiments. Live litters containing KI/KI mice were produced from intercrossing *tgfb1*<sup>R278A</sup>/*tgfb1*<sup>R278A</sup> or *tgfb1*<sup>R278A</sup>/*tgfb1*<sup>WT</sup> mice, or crossing *tgfb1*<sup>R278A</sup>/*tgfb1*<sup>R278A</sup> to *tgfb1*<sup>R278A</sup>/*tgfb1*<sup>WT</sup>. To confirm mutant mRNA production from the KI and knock-out alleles, RNA was extracted from tail clippings, cDNA synthesized and amplified using the respective primer pairs *tgfb1* Ex3/4 cDNA F and *tgfb1* ex 6/7 cDNA R, and *tgfb1* Ex1/2 cDNA F and *tgfb1* Ex4/5 cDNA R and the products sequenced. To confirm the absence of mature TGF-β1 protein in the KO and absence of released mature TGF-β1 in the KI mice, immunoblots were performed using an antibody to mouse mature TGF-β1 (Abcam, ab179695). Serum was precleared 3 times with Protein G Sepharose beads to deplete IgG prior to immunoblotting. Spectral flow cytometry was performed on peripheral blood, or spleen from *tgfb1*<sup>-/-</sup>, KI/KI (*tgfb1*<sup>R278A</sup>/*tgfb1*<sup>R278A</sup>) or appropriate age and littermate matched controls (WT/WT, WT/KO or WT/KI). Histologic analysis of various organs were scored on an inflammation scale of 0–3 (0 = no inflammation; 1 = scattered lymphocytes infiltrating into tissues; 2 = distinct aggregates of lymphocytes infiltrating into tissues; 3 = diffuse inflammation infiltrating tissue in dense sheets of lymphocytes) (Table S1). Total inflammation score represents the sum of all individual organ inflammation scores.

### Cell lines

Transformed mink lung TGF-β reporter cells (TMLC)<sup>47</sup> were a gift from J. Munger (New York University Medical Center, New York, NY, USA) and were stably transfected with L-TGF-β1 (RGD/RGD), L-TGF-β1 (RGD/RGE), L-TGF-β1 (RGE/RGE), L-TGF-β3 (RGD/RGD), L-TGF-β3\_lasso3 with or without GARP, as previously described.<sup>22</sup> TMLC cells were grown in DMEM + 10% FBS + penicillin-streptomycin + amphotericin B, cultured at 37°C in a humidified incubator, 5% CO<sub>2</sub>.

MFB-F11 cells were a gift from Tony Wyss-Coray (Stanford University School of Medicine). MFB are a mouse fibroblast line from *tgfb1*<sup>-/-</sup> mice which were stably transfected with an SBE-SEAP reporter cassette with a hygromycin resistance cassette and clone F11 isolated by limiting dilution.<sup>54</sup> MFB-F11 cells were stably transduced with human *ITGB8* construct using retroviral particles from the Phoenix amphotropic viral packaging cell line (Phoenix-AMPHO, ATCC). MFB-F11 cells were maintained in DMEM + 10% FBS + penicillin-streptomycin + amphotericin B, cultured at 37°C in a humidified incubator, 5% CO<sub>2</sub>. β8 expression was maintained by supplementing basal media with 5 μg/mL puromycin. Phoenix cells were maintained in DMEM + 10% FBS + penicillin-streptomycin + amphotericin B, cultured at 37°C in a humidified incubator, 5% CO<sub>2</sub>.

### DNA constructs

The following cDNA constructs were used: β8 cDNA pBABE puro, αvfl pcDM8, αvtr pcDM8, β8tr pcDNA1neo, β8fl pcDNA1neo<sup>22,28</sup>; pLX307 hTGF-β1 IRES2 EGFP (h preceding protein name indicates human from here forward) was constructed from TGF-β1\_pLX307 (Plasmid #98377, AddGene) to remove a c-terminal V5 tag by cloning a PCR fragment created with primers (5'-caggtgtcgtgagggc tagcatcg-3', and 5'-gcgccactagtctcgagttatcag-3') which was used as a backbone to generate pLX307 hTGF-β1 RGE\_IRES2 EGFP puro, pLX307 hTGF-β1 RGD\_R249A\_IRES2 EGFP puro, pLX307 hTGF-β1 RGE\_R249A\_IRES2 EGFP puro, as described,<sup>22</sup> L-TGF-β1\_RGD\_Lasso3 (where the A31-L44 in lasso1 loop was swapped with T31-V42 from the L-TGF-β3 lasso3 loop) was made by splice overlap extension PCR using the primers (5'-ccatttcaggtgtcgtgagggc-3', 5'-ccctgagccaacggtgatgccaccgtcccc gagggcgtgtctgcg-3', 5'-gtcatcaccgttggtcagggggcgtgtgagccgcagcttgacag-3', 5'-tgccgtagtagtcgagcctc-3'). HA-GARP pcDNA3<sup>78</sup> was a gift from Sophie Lucas (Institut de Duve, Belgium), HIS SBP human GARP (hGARP) pcDNA6 was made using an N-terminal rat albumin signal peptide-His Tag-Streptavidin binding protein-HRV 3C protease tag (HIS SBP) from HIS SBP tagged porcine L-TGF-β1

pcDNA6<sup>22</sup> as a template using primers (5'-ctctgatatcccaagctggctagccacc-3', 5'-cagggcactttgtcttggtgaggaccctgaaacagcacctc-3') and joined by splice-overlap extension to a fragment amplified from HA GARP pcDNA6<sup>95</sup> using primers (5'-tagaggtgctgtttca gggctctcaccagaacaaagtgccctg-3', and 5'-ccgctgtacaggtgttccc-3'), HIS SBP hGARP tr pcDNA6 was made by ligation of a PCR amplified fragment (HIS SBP hGARP fl pcDNA6 as a template with the primers 5'-agggccgtgtggacgtgg, and 5'-tctcctcgattatcagtt gatgttctcagtcaccccttc-3'), HIS SBP hGARP tr pcDNA6 SpyCatcher was generated from HIS SBP-GARP tr pcDNA6 by gapping into the XhoI/XbaI cut plasmid a PCR fragment amplified from addgene-plasmid-133447 (SpyCatcher) using primers (5'- ggggactgaa gaacatcaacatgctgactaccatcaccatc-3'; 5'- ggcttaccttgaaggcccttagctaccactggatccagta-3') using the Gibson Assembly Cloning Kit (NEB #E5510S). The entire human open reading frame and IRES RED cassette was transferred using PmeI/Spel from pLVE-HTGFB3-IRES-RED (Plasmid #52580, addgene) to replace the TGF- $\beta$ 1 reading frame and IRES GFP (Clal/Klenow, SpeI) into TGF- $\beta$ 1\_pLX307 to create hTGF- $\beta$ 3 IRES RED, hTGF- $\beta$ 3 R277A IRES RED was made using splice overlap extension using PCR products (5'-ccatgtcacaccttcagccc-3', 5'-gtccaaagccgccttcttctctg-3'; 5'-cagaggaagaaggcggcgtttggac-3', gttgtgtacagtcaccagcacc), hTGF- $\beta$ 3 RGE\_R277A\_IRES RED was made using splice overlap extension using PCR products (5'- ggcgccccagttctccacgg-3', 5'-ggcgccccagttctccacgg-3'; 5'-ggagaactggggcgccctcaag-3', 5'-gtccaaagccgccttcttctctg-3'; 5'-cagaggaagaaggcggcgtttggac-3', 5'-gtgtgtacagtcaccagcacc-3'), hTGF- $\beta$ 3 RGE\_IRES RED was made using splice overlap extension using PCR products (5'- ggcgc cccagttctccacgg-3', 5'- gtccaaagccgccttcttctctg-3'; 5'- ggagaactggggcgccctcaag-3', 5'-gtgtgtacagtcaccagcacc-3'), hTGF- $\beta$ 3 IRES GFP, hTGF- $\beta$ 3 RGE\_IRES GFP, hTGF- $\beta$ 3 R277A\_IRES GFP and hTGF- $\beta$ 3 RGE\_R277A\_IRES GFP were made by cloning in the PCR fragment generated using TGF- $\beta$ 1\_pLX307 as a template (5'- ctctacgcgtactagtgccgcgcgg-3', 5'-ttactgtacagtcgtc catgcc-3') and cloning into hTGF- $\beta$ 3 IRES RED, hTGF- $\beta$ 3 R277A\_IRES RED, hTGF- $\beta$ 3 RGE\_IRES RED and hTGF- $\beta$ 3 RGE\_R277A\_IRES RED, SBP HIS L-TGF- $\beta$ 3, SBP HIS L-TGF- $\beta$ 3 RGE, SBP HIS L-TGF- $\beta$ 3 R277A, SBP HIS L-TGF- $\beta$ 3 R277A\_RGE, SBP HIS L-TGF- $\beta$ 3 C4S, SBP HIS L-TGF- $\beta$ 3 C4S RGE, SBP HIS L-TGF- $\beta$ 3 C4S R277A, and SBP HIS L-TGF- $\beta$ 3 C4S R277A\_RGE all in pcDNA6 were made using splice overlap extension PCR to amplify the rat albumin signal peptide-HIS SBP-HRV 3C protease tag from HIS SBP L-TGF- $\beta$ 3 pcDNA6 using primers specific for HIS SBP L-TGF- $\beta$ 3 C4WT (5'-gactcactatagg gagaccaagctgg-3', 5'- gtccaaggtggtgcaagtgacagggacctgaaac-3'; 5'-ctgtccacttgaccaccttgac-3', 5'- ggtgagcctaagcttgctca agatctg-3') or HIS SBP L-TGF- $\beta$ 3 C4S: 5'- gactcactatagggagaccaagctgg-3', 5'- gtccaaggtggtgctagtgacagggacctgaaac-3'; 5'- ctgtccactagaccaccttgac-3', 5'- ggtgagcctaagcttgctcaagatctg-3'), and ligating the corresponding spliced products into hTGF- $\beta$ 3 IRES GFP, hTGF- $\beta$ 3 RGE\_IRES GFP, hTGF- $\beta$ 3 R277A\_IRES GFP and hTGF- $\beta$ 3 RGE\_R277A\_IRES GFP. All cDNA constructs were verified by sequencing. TGF- $\beta$ R2-Fc was previously described<sup>27</sup>.

## METHOD DETAILS

### Antibody isolation, characterization, and production

The following antibodies were used: and anti- $\beta$ 8 clone F12, which is high-affinity derivative of the parental clone C6D4<sup>80</sup> created by combining rational structure based directed evolution to create a mutagenic Vh and VI domain library focused on amino acids to optimize the binding interface, displayed on the surface of yeast, and after multiple rounds of sorting, isolation and subcloning into murine IgG2a format, as described,<sup>80</sup> produced in ExpiCHO cells.

### Expression and purification of proteins for functional assays and single particle cryo-EM

Secreted ectodomain of  $\alpha$ v $\beta$ 8 integrin was produced by transfecting ExpiCHO cells with integrin constructs<sup>22</sup> using the manufacturer's protocol. Specifically, after 5 days growth, cell culture was centrifuged to collect supernatant, which was filtered through a PES (polyether sulfone) membrane, 0.2  $\mu$ m pore size (Millipore). Protein purification is carried out by affinity chromatography using a column packed with Protein G crosslinked by antibody 8B8 which binds to  $\alpha$ v integrin.<sup>96</sup> Bound  $\alpha$ v $\beta$ 8 is eluted from beads by washing the column with 100 mM glycine at pH 2.5. Flow through is immediately buffer adjusted by 2 M Tris-HCl pH 8, followed by size exclusion chromatography (Superose 6 Increase 10/300 GL, GE Healthcare) in 20 mM Tris-HCl pH 7.4, 150 mM NaCl, 1 mM CaCl<sub>2</sub> and 1 mM MgCl<sub>2</sub>.

Full length of  $\alpha$ v $\beta$ 8 integrin was produced by transfecting ExpiCHO cells with integrin constructs using the manufacturer's protocol. Cells were harvested after 3 days growth. Cells were solubilized by rotation in 4 °C using solubilize buffer for 3 hrs (20 mM HEPES, pH 8.0, 150 mM NaCl, 1 mM CaCl<sub>2</sub>, 1 mM MgCl<sub>2</sub>, 10 mM DDM, 2 mM CHS and 2% OG, 1x Protease Inhibitor Cocktail, EDTA-Free). Supernatant containing proteins were collected by centrifuged at 4,000 g followed by ultra-speed centrifuge at 45,000 rpm. Protein purification is carried out by affinity chromatography using a column packed with Protein G crosslinked by antibody C6D4F12 which binds to  $\alpha$ v $\beta$ 8 integrin. Bound full length  $\alpha$ v $\beta$ 8 is eluted from beads by washing the column with elution buffer (100 mM glycine at pH 2.5, 0.03% DDM). Flow through is immediately buffer adjusted by 2 M Tris-HCl pH 8.0, followed by size exclusion chromatography (Superose 6 Increase 10/300 GL, GE Healthcare) in 20 mM Tris-HCl pH 7.4, 150 mM NaCl, 0.03% DDM, 1 mM CaCl<sub>2</sub> and 1 mM MgCl<sub>2</sub>.  $\alpha$ v $\beta$ 8 in nanodisc was made by adding at a ratio of  $\alpha$ v $\beta$ 8fl: MSP-2N2: lipid equals to 1: 4: 200 in 4 °C for 3 hrs, biobeads were added to remove the residue lipids over night by gentle rotation.  $\alpha$ v $\beta$ 8 in nanodisc was collected and further purified by size exclusion chromatography (Superose 6 Increase 10/300 GL, GE Healthcare) in 20 mM Tris-HCl pH 7.4, 150 mM NaCl, 1 mM CaCl<sub>2</sub> and 1 mM MgCl<sub>2</sub>, the pooled and concentrated protein was subjected to SDS-PAGE, each protein size was identified to be corrected (Figure S4F).



Similarly, secreted L-TGF- $\beta$ 1/GARP was produced by transient transfecting Expi293 cells with three different constructs, L-TGF- $\beta$ 1 with R249A mutation, L-TGF- $\beta$ 1 with R249A and RGE mutation, and ectodomain of GARP with N-terminal Strep-His tag. This strategy favors formation of L-TGF- $\beta$ 1/GARP with a single intact RGD integrin binding motif. Cell culture was centrifuged to collect supernatant, which was filtered through a PES (polyether sulfone) membrane, 0.2  $\mu$ m pore size (Millipore). Protein purification is carried out by using Ni-NTA agarose (QIAGEN), washed with three column volumes of 0.6 M NaCl, 0.01 M Tris (pH 8.0) and eluted with 0.25 M imidazole in Tris-buffered saline (TBS). The elution was then applied to Strep-tactin agarose (IBA) and washed with TBS (pH 7.4). To cleave the tag, 3.5  $\mu$ l of commercial HRV-3C protease (Novagen, 1.8–3.0 U/ $\mu$ l) in TBS (pH 7.4) with 20% glycerol, was applied to the column, and incubated at 4°C overnight. The flow-through was washed with two column volumes of TBS (pH 7.4), then concentrated using centrifugal concentrators (Millipore) to about 1 mg/ml in 10 mM Tris (pH 7.4), 150 mM NaCl.

L-TGF- $\beta$ 3 was produced by transiently transfecting 293T cells with equal amounts of human L-TGF- $\beta$ 3 C4S\_R277A\_RGD and C4S\_R277A\_RGE plasmids.

C6D4F12 was produced by co-transfecting F12 VH pcDNA3.1 and F12 VL pcDNA3.1 into ExpiCHO cells and antibody purified using protein G agarose, as described.<sup>22</sup>

The homogeneity and purity of all protein preparations were verified by SDS-PAGE stained with Coomassie blue and protein concentrations were measured by nanodrop.

### Mass photometry

Mass photometry experiments were performed with a Refeyn OneMP (Refeyn Ltd.). Each sample in TBS buffer with 1 mM CaCl<sub>2</sub>, 1 mM MgCl<sub>2</sub> of 16  $\mu$ l was pipetted into the reaction chambers. Calibration was carried out by BSA, apoferritin and ADH. L-TGF- $\beta$ 1/GARP, L-TGF- $\beta$ 3/GARP,  $\alpha$ v $\beta$ 8,  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP, and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3/GARP sample were diluted to 0.1 mg/ml, 1  $\mu$ l of each sample was added to a 15  $\mu$ l TBS with 1 mM CaCl<sub>2</sub> and 1 mM MgCl<sub>2</sub> buffer already pipetted into the reaction chamber. Image analysis was performed and analyzed by the software provided by Refeyn Ltd., with the default settings provided by the manufacturer.

### Cryo-EM sample preparation

We co-expressed the recombinant GARP ectodomain, L-TGF- $\beta$ 1 with a wild type integrin binding motif RGD (L-TGF(RGD)- $\beta$ 1), and a mutant form (L-TGF(RGE)- $\beta$ 1) that cannot bind to integrin. The resulting purified L-TGF- $\beta$ 1/GARP contains about 50% L-TGF- $\beta$ (RGE/RGD)- $\beta$ 1, which can only bind one  $\alpha$ v $\beta$ 8 integrin, 25% L-TGF(RGD/RGD)- $\beta$ 1, which can bind two, and 25% L-TGF(RGE/RGE)- $\beta$ 1, which cannot bind to  $\alpha$ v $\beta$ 8. This design allows us to maximize the population of L-TGF- $\beta$ /GARP bound with one  $\alpha$ v $\beta$ 8, reducing heterogeneity of the sample and facilitates particle alignment.<sup>22</sup> Such L-TGF- $\beta$ 1/GARP was mixed with  $\alpha$ v $\beta$ 8 in 1:1 molar ratio and incubated at room temperature for 30 min, the final protein complex concentration is 0.5 mg/ml. For cryo-EM grid preparation, 3  $\mu$ l of the complex was deposited onto QUANTIFOIL® R 1.2/1.3 on Au 300 mesh grids and UltrAuFoil® R 1.2/1.3 on Au 300 mesh grids. Grids were pre-glow-discharged for 30 s at 15 mA prior to sample application and freezing. The complexes were frozen using a FEI Vitrobot Mark IV using a 1 s blot time with blot force 1. All grids were frozen with 100% humidity at 22°C and plunge-frozen in liquid ethane cooled by liquid nitrogen.

L-TGF- $\beta$ 1/GARP-SpyCatcher was mixed with  $\alpha$ v $\beta$ 8 in 1:1 molar ratio and incubated at room temperature for 30 mins, the final protein complex concentration is 0.15 mg/ml. For cryo-EM grid preparation, 3  $\mu$ l of the complex was deposited onto UltrAuFoil® R 1.2/1.3 on Au 300 mesh grids, covered with graphene oxide functionalized by Spy-tag,<sup>97</sup> washed by 10  $\mu$ l TBS buffer 3 times, finally 3  $\mu$ l TBS buffer was added. The complexes were frozen using a FEI Vitrobot Mark IV using a 3 s blot time, with 100% humidity at 22°C and plunge-frozen in liquid ethane cooled by liquid nitrogen.

L-TGF- $\beta$ 1/GARP mixed with  $\alpha$ v $\beta$ 8fl-nd in 1:1 molar ratio and incubated at room temperature for 30 min, the final protein complex concentration is 0.5 mg/ml. For cryo-EM grid preparation, 3  $\mu$ l of the complex was deposited onto QUANTIFOIL® R 1.2/1.3 on Au 300 mesh grids. Grids were glow-discharged for 30 s at 15 mA prior to sample application and freezing. The complexes were frozen using a FEI Vitrobot Mark IV using a 1 s blot time. All grids were frozen with 100% humidity at 22°C and plunge-frozen in liquid ethane cooled by liquid nitrogen.

To prepare the complex of L-TGF- $\beta$ 3/GARP with  $\alpha$ v $\beta$ 8, the molar ratio was 1:1, and incubated at room temperature for 30 min, the final protein complex concentration is 0.5 mg/ml. For cryo-EM grid preparation, 3  $\mu$ l of the complex was deposited onto QUANTIFOIL® R 1.2/1.3 on Au 300 mesh grids, grids were glow-discharged for 30 s at 15 mA prior to sample application and freezing. The complexes were frozen using a FEI Vitrobot Mark IV using a 1 s blot time. All grids were frozen with 100% humidity at 22°C and plunge-frozen in liquid ethane cooled by liquid nitrogen.

To prepare the complex of L-TGF- $\beta$ 1/GARP and L-TGF- $\beta$ 3/GARP, the concentration is 0.3 mg/ml. For cryo-EM grid preparation, 3  $\mu$ l of the complex was deposited onto QUANTIFOIL® R 1.2/1.3 on Au 300 mesh grids, grids were glow-discharged for 30 s at 15 mA prior to sample application and freezing. The complexes were frozen using a FEI Vitrobot Mark IV using a 1 s blot time. All grids were frozen with 100% humidity at 8°C and plunge-frozen in liquid ethane cooled by liquid nitrogen.

To prepare the complex of L-TGF- $\beta$ 3 with  $\alpha$ v $\beta$ 8, 100  $\mu$ g of recombinant  $\alpha$ v $\beta$ 8 was incubated 150  $\mu$ g L-TGF- $\beta$ 3, incubated at room temperature for 30 min, subjected to size exclusion chromatography and concentrated to 0.45 mg/ml. For cryo-EM grid preparation, 2.5  $\mu$ l of the complex was deposited onto Quantifoil grids. 0.075 mg/ml sample was onto 400 mesh 1.2/1.3 copper Quantifoil Graphene-oxide grid, 0.05 mg/ml sample was onto 400 mesh 1.2/1.3 Au Quantifoil Graphene-oxide grid. 0.25 mg/ml sample was onto

400 mesh R 1.2/1.3 copper Quantifoil grid and 400 mesh R 1.2/1.3 Au Quantifoil grid. Except for the Graphene oxide grid, grids were glow-discharged for 60 s at 15 mA prior to sample application and freezing. The Graphene oxide grids were frozen using a FEI Vitrobot Mark IV using a 6 s blot time, the rest used 4 s. All grids were frozen with 100% humidity at 20°C and plunge-frozen in liquid ethane cooled by liquid nitrogen.

### Cryo-EM data acquisition

All the automated data collections below were carried out using the SerialEM.<sup>87</sup> For L-TGF- $\beta$ 1/GARP, the data set was collected on a Thermo Fisher 300 KeV Titan Krios G2 equipped with a GATAN K3 direct detector camera. 2,939 movies were collected at a nominal magnification of 105,000x, the defocus range was set to be between -1.1 and -2.2  $\mu\text{m}$ . The detector pixel size was 0.834 Å and the dose was 46  $\text{e}^-/\text{Å}^2$ .

For  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP and  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP-SpyCatcher, the data set was collected on a Thermo Fisher 300 KeV Titan Krios G2 equipped with a GATAN K3 direct detector camera. 18,819 movies were collected at a nominal magnification of 105,000x, the defocus range was set to be between -0.8 and -2.5  $\mu\text{m}$ . The detector pixel size was 0.835 Å and the dose was 68  $\text{e}^-/\text{Å}^2$ . In the data set, 4176 movies were collected on QUANTIFOIL® R 1.2/1.3 on Au 300 mesh grids, 7,777 movies were collected on UltrAuFoil® R 1.2/1.3 on Au 300 mesh grids, 1,867 movies were collected on UltrAuFoil® R 1.2/1.3 on Au 300 mesh grids by tilting 30°, 4,999 movies were collected on  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP-SpyCatcher.

For  $\alpha\text{v}\beta$ 8fl-nd, the data set was collected on a ThermoFisher 300 KeV Titan Krios G2 equipped with a GATAN K3 direct detector camera. 17,550 movies were collected at a nominal magnification of 105,000x, the defocus range was set to be between -0.8 and -2.5  $\mu\text{m}$ . The detector pixel size was 0.8189 Å and the dose was 47.4  $\text{e}^-/\text{Å}^2$ .

For L-TGF- $\beta$ 3/GARP, the data set was collected on a Thermo Fisher 300 KeV Titan Krios G2 equipped with a GATAN K3 direct detector camera. 3,666 movies were collected at a nominal magnification of 130,000x, the defocus range was set to be between -1.1 and -2.2  $\mu\text{m}$ . The detector pixel size was 0.664 Å and the dose was 47  $\text{e}^-/\text{Å}^2$ .

For  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 3/GARP, the data set was collected on a Thermo Fisher 200 KeV Talos Arctica equipped with a GATAN K3 direct detector camera. 1,654 movies were collected at a nominal magnification of 28,000x, the defocus range was set to be between -1.1 and -2.2  $\mu\text{m}$ . The detector pixel size was 1.430 Å and the dose was 61  $\text{e}^-/\text{Å}^2$ .

For  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 3, the datasets were acquired on a Thermo Fisher 300 KeV Titan Krios G2 operated in nano-probe mode at 300 kV equipped with a Gatan Quantum GIF energy filter operated in zero-loss mode with a slit width of 20 eV and a Gatan K2 Summit direct detector. Movies were recorded in super resolution mode with a super resolution pixel size of 0.6725 Å/pix and a nominal magnification of 105 kx at a dose rate of 7.7  $\text{e}^-/\text{pix}/\text{s}$ . Each 16 s movies contained 80 frames of 200 ms each, which corresponds to a total dose of  $\sim 70 \text{ e}^-/\text{Å}^2$ , collected in a single session with a nominal defocus range of 1.2 - 2.4  $\mu\text{m}$  under focus. 400 mesh 1.2/1.3 copper Quantifoil Graphene-oxide grid collected 878 movies, 400 mesh 1.2/1.3 Au Quantifoil Graphene-oxide grid collected 747 movies, 400 mesh R 1.2/1.3 copper Quantifoil grid collected 234 movies, 400 mesh R 1.2/1.3 Au Quantifoil grid collected 2,237 movies.

### Imaging processing

For all individual datasets, dose fractionated super-resolution image stacks were motion corrected and binned 2 by Fourier cropping using MotionCor2.<sup>85</sup> The entire data processing and map reconstruction was carried out with cryoSPARC.<sup>88</sup> CTF estimation was performed by patch CTF module in cryoSPARC.

Cryo-EM dataset of  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP-SpyCatcher with graphene oxide functionalized by Spy-tag grid was collected as a control experiment to rule out the influence of air-water interface on reconstruction. Reconstruction from this dataset show the same structural features as the one from  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP complex. We therefore combined these two datasets together. All micrographs collected from  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP and  $\alpha\text{v}\beta$ 8/L-TGF- $\beta$ 1/GARP-SpyCatcher are combined. Initial particle picking carried out by multi template picking (L-TGF- $\beta$ 1/GARP with one or two integrins) identified 6,438,291 particles (Figures S3C–S3G). After multiple rounds of 2D classifications to eliminate duplicate particles and obvious junk, followed by Ab-Initio Reconstruction combined with Heterogeneous Refinement procedures in cryoSPARC, two major classes were identified, i.e., L-TGF- $\beta$ 1/GARP bound with two integrins (643,335 particles) and one integrin (1,324,888 particles). Further NU-refinement of the first class produced a reconstruction of L-TGF- $\beta$ 1/GARP bound with two integrins at a nominal resolution of 3.21 Å. This map was further improved by DeepEnhancer.<sup>98</sup>

NU-refinement of the second class produced a reconstruction of L-TGF- $\beta$ 1/GARP bound with one integrin at a nominal resolution of 2.54 Å. To resolve GARP, all particles in this class were subjected to two more rounds of Ab-Initio Reconstruction combined with Heterogeneous Refinement, which produced four classes. One class (94,847 particles) shows a relatively stable conformation of L-TGF- $\beta$ 1/GARP on top of  $\alpha\text{v}\beta$ 8, with a minor density of a second  $\alpha\text{v}\beta$ 8 binding. This class was subjected to further 3D classification (Beta), in cryoSPARC, in which, a sphere mask on GARP is applied. This procedure isolated 45,913 particles with GARP in an up conformation, 30,190 particles with GARP in a down conformation, with the remaining ambiguous. Particles of both classes were further classified by an additional round of Ab-Initio Reconstruction combined with heterogeneous refinement to remove remaining particles bound with two  $\alpha\text{v}\beta$ 8. For the class of GARP facing down and one integrin bound (15,748 particles), further refinement by NU-refinement produced a 4.12 Å resolution reconstruction. For the class of GARP facing up and one integrin bound (22,815 particles), one more round of Ab-Initio reconstruction combined with heterogeneous refinement produced a reconstruction

of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 1/\text{GARP}$  with GARP facing up at 4.05 Å. For particles with GARP in the down conformation, a similar procedure was applied to produce a final reconstruction of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 1/\text{GARP}$  at 4.11 Å resolution.

All discarded particles during the above processes were pooled and subjected to multiple rounds of Ab-Initio Reconstruction combined with Heterogeneous Refinement, to rescue all particles that belong to the GARP in up conformation with one  $\alpha\text{v}\beta 8$  bound. Merging the rescued particles with the existing particles from well-resolved classes produced a total of 46,771 particles, that were used for a final round NU-refinement, producing a final reconstruction of L-TGF- $\beta 1$  with GARP in the up position and bound with one  $\alpha\text{v}\beta 8$  at a nominal resolution of 3.91 Å. Further local refinement with masks on either L-TGF- $\beta 1/\text{GARP}$  or  $\alpha\text{v}\beta 8$  produced two final reconstructions with nominal resolutions of 3.3 Å, and 3.1 Å, respectively. Because two masks overlap with each other, these two reconstructions were merged by joining the common region into a final combined map. A focused 3DVA was performed on this final particle stack by applying a mask on L-TGF- $\beta 1/\text{GARP}$  to reveal the motion of this part of the reconstruction. By random splitting this 46,771 particles data to 3 parts, each part was performed NU-refinement, producing a final reconstruction. Further local refinement with masks on either L-TGF- $\beta 1/\text{GARP}$  or  $\alpha\text{v}\beta 8$  produced two final reconstructions. Local resolutions were determined and compare with the full data set of 46,771 particles.

The class with 1,324,888 particles of L-TGF- $\beta 1/\text{GARP}$  with one  $\alpha\text{v}\beta 8$  bound underwent further guided classification by multiple rounds of Ab-Initio Reconstruction combined with Heterogeneous Refinement, using a map of L-TGF- $\beta 1/\text{GARP}$  as a reference. This isolates 34,117 particles of L-TGF- $\beta 1/\text{GARP}$  alone not bound with  $\alpha\text{v}\beta 8$ . NU-refinement result of this subset produced a reconstruction of 3.47 Å, to which we applied DeepEmhancer procedure to improve structural features.

All particles that produced the two clear reconstructions described above as well as the reconstruction of L-TGF- $\beta 1/\text{GARP}$  alone were removed from the 1,324,888 particles dataset, and remaining particles were transferred to Relion 3.8<sup>99</sup> for focused classification without alignment with a sphere mask applied on L-TGF- $\beta 1/\text{GARP}$ . Each good class was transferred back to cryoSPARC for NU-refinement. Among them is a class that is  $\alpha\text{v}\beta 8$  alone, at a resolution of 4.54 Å. Particles of all good classes with one  $\alpha\text{v}\beta 8$  were pooled and subjected to a 3DVA masked on L-TGF- $\beta 1/\text{GARP}$  and partially on  $\alpha\text{v}\beta 8$ .

The data processing of L-TGF- $\beta 1/\text{GARP}$  is shown in Figure S3H. The entire data processing and map reconstruction was carried out with cryoSPARC. The initial particle picking identified 1,845,468 particles. After several rounds of 2D classifications, Ab-Initio Reconstruction combined with Heterogeneous Refinement, about 318,954 particles were isolated for NU-refinement, yielding a map with a nominal resolution of 3.0 Å. By random splitting this data to 3 parts, each part was performed NU-refinement, producing a final reconstruction. Local resolutions were determined and compare with the full data.

The data processing of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 1/\text{GARP}/\text{MHG8-Fab}$  is shown in Figure S4E. The entire data processing and map reconstruction was carried out with cryoSPARC. The initial particle picking identified 292,348 particles. After several rounds of 2D classifications, Ab-Initio Reconstruction combined with Heterogeneous Refinement, about 16,774 particles were isolated for NU-refinement, yielding a map with a nominal resolution of 7.72 Å. Further classifications were applied, showing not much density could be resolved well on  $\alpha\text{v}\beta 8$ .

Data processing of  $\alpha\text{v}\beta 8\text{fl-nd}/\text{L-TGF-}\beta 1/\text{GARP}$  is shown in Figure S4G. The entire data processing and map reconstruction was carried out with cryoSPARC. The initial particles picking identified 2,335,916 particles. After multi rounds of 2D classifications, Ab-Initio Reconstruction combined with Heterogeneous Refinement, about 20,547 particles were isolated for NU-refinement, yielding a map with a nominal resolution of 3.45 Å. Further local refinement with masks on either L-TGF- $\beta 1/\text{GARP}$  or  $\alpha\text{v}\beta 8$  produced two final reconstructions with nominal resolutions of 6.3 Å, and 3.1 Å, respectively. Because two masks overlap with each other, these two reconstructions were merged by joining the common region into a final combined map.

The data processing of L-TGF- $\beta 3/\text{GARP}$  is shown in Figure S5A. The entire data processing and map reconstruction was carried out with cryoSPARC. The initial particle picking identified 522,776 particles. After several rounds of 2D classification followed by Ab-Initio Reconstruction combined with Heterogeneous Refinement, about 151,804 particles were isolated for NU-refinement, yielding a nominal resolution of 2.93 Å.

The data processing of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 3/\text{GARP}$  is shown in Figure S5E. The entire data processing and map reconstruction was carried out with cryoSPARC. The initial particle picking identified 1,007,086 particles. After several rounds of 2D classification, Ab-Initio Reconstruction combined with Heterogeneous Refinement yielded about 151,804 suitable particles which were used to separate into 3 classes, with L-TGF- $\beta 3/\text{GARP}$  in distinct conformations, and NU-refinement was performed for each class, yielding nominal resolutions of 5.89 Å, 4.86 Å and 7.19 Å.

Data processing of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 3$  is shown in Figure S5F. The entire data processing and map reconstruction was carried out with cryoSPARC. The initial particle picking identified 779,392 particles. After several rounds of 2D classifications, Ab-Initio Reconstruction combined with Heterogeneous Refinement, about 382,107 particles were isolated and used for NU-refinement, yielding a nominal resolution of 2.73 Å. In the intermediate data set, 537,399 particles were identified and extracted to Relion 3.8, masked on L-TGF- $\beta 3$  for 3D classification without alignment, and each class transferred back to cryoSPARC for NU-refinement.

### Model building and refinement

For atomic model building of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 1/\text{GARP}$ , the atomic models of  $\alpha\text{v}\beta 8$  (PDB: 6UJA)<sup>22</sup> and L-TGF- $\beta 1/\text{GARP}/\text{MHG8}$  crystal structure (PDB: 6GFF)<sup>9</sup> were used as an initial model. For atomic model building of L-TGF- $\beta 1/\text{GARP}$ , atomic model of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 1/\text{GARP}$  with  $\alpha\text{v}\beta 8$  portion removed was used as an initial model. For atomic model building of L-TGF- $\beta 3/\text{GARP}$ , a homology model

predicted by Swiss-model<sup>100</sup> based on crystal structure (PDB: 6GFF) was used as the initial model. For atomic model building of  $\alpha\text{v}\beta 8/\text{L-TGF-}\beta 3$ , the atomic models of  $\alpha\text{v}\beta 8$  (PDB: 6UJA)<sup>22</sup> and crystal structure (PDB: 4UM9)<sup>101</sup> were used as initial models.

In each case, the initial model was fitted as a rigid body into corresponding cryo-EM map by UCSF Chimera followed by Phenix real\_space\_refinement.<sup>102</sup> ISOLDE<sup>103</sup> in UCSF ChimeraX<sup>100</sup> was used to build disulfide and to refine the side chain. The model was checked in COOT<sup>104</sup> with some side chains manually adjusted to fit the density, followed by a final real space refinement in Phenix with a B-factor refined for each residue.

### B-factor comparison

For L-TGF- $\beta 1/\text{GARP}$ ,  $\alpha\text{v}\beta 8\text{tr}/\text{L-TGF-}\beta 1/\text{GARP}$ , and L-TGF- $\beta 3/\text{GARP}$ , each atomic model, B-factors of individual residues were refined during the final real space refinement in Phenix.<sup>102</sup> In all structures, except  $\alpha\text{v}\beta 8\text{fl-nd}$ , L180 in GARP is relative stable and has the relative lower B-factor, thus was chosen as a common reference point with its B-factor set to 0Å.<sup>2</sup> The B-factor of every other residue in the structure was normalized by the difference of its refined B-factor with that of L180.

For  $\alpha\text{v}\beta 8\text{tr}/\text{L-TGF-}\beta 1/\text{GARP}$  and  $\alpha\text{v}\beta 8\text{fl-nd}$ , some relative stable residues (L106, L171, V228, A281, Y374, M408) in  $\alpha\text{v}$  subunit were averaged in each model, and chosen to be normalized by division between two models.

The difference of B-factor of individual residue between two structures was measure by the difference of the normalized B-factor. B-factor coloring was created by UCSF ChimeraX.<sup>105</sup> Because of the sequence differences between L-TGF- $\beta 1$  and L-TGF- $\beta 3$ , the B-factor differences between the two were measured by the difference of the normalized B-factor assigned to C $\alpha$  atoms. B-factor C $\alpha$  coloring was created by UCSF ChimeraX.<sup>105</sup>

### RMSD measurement

The Root-Mean-Square Deviation (RMSD) between L-TGF- $\beta 1/\text{GARP}$  and L-TGF- $\beta 1/\text{GARP}/\alpha\text{v}\beta 8$  was measure by the command *rmsd sel* in Chimera, only the shared resolved residues were selected for each subunit. RMSD between GARP is 1.336 Å, RMSD between non-integrin binding subunit of L-TGF- $\beta 1$  is 1.934 Å, RMSD between integrin binding subunit of L-TGF- $\beta 1$  is 2.034 Å.

### Molecular dynamics simulations

The cryo-EM structure of the L-TGF- $\beta 1/\text{GARP}$  complex was used as a reference starting structure for the molecular dynamic simulations. The unresolved loops were modeled using Rosetta protocols.<sup>106</sup> The GARP  $\beta$ -hairpin loop, I271 to A280, was added based on homology modeling with the X-ray crystal structure (PDB: 6GFF).<sup>9</sup> The CHARMM36m<sup>107</sup> force field was used for the protein and ions. The CHARMM-modified TIP3P forcefield<sup>108</sup> was used for the water molecules. The L-TGF- $\beta 1/\text{GARP}$  complex was inserted in a box of dimensions 150.4 Å x 150.7 Å x 150.9 Å and solvated with 99,086 water molecules. This arrangement ensured that each protein atom was at least 20 Å away from its periodic image. To neutralize the charge of the entire system, 284 Na<sup>+</sup> and 281 Cl<sup>-</sup> ions were added. The final system comprised a total of 318,045 atoms. The molecular dynamics simulations were performed with GROMACS 2023.<sup>109</sup> We employed periodic boundary conditions and calculated nonbonding interactions using the particle mesh Ewald method<sup>110</sup> with a 12Å cutoff. The bonds involving hydrogen atoms were restrained using the LINCS algorithm.<sup>111</sup> We used the Nose-Hoover thermostat and Parrinello-Rahman Barostat. First, an NVT equilibration for 125 ps at 303 K was performed with a friction coefficient of 1 ps<sup>-1</sup> and timestep of 1fs. Then, the timestep was set to 2 fs and 10 ns of an NPT equilibration was performed until the average pressure of the system was equilibrated to 1 atm at 303 K. The final frame of the NPT equilibration was extracted to initiate two production runs, each 1  $\mu\text{s}$  long. Conformational snapshots were saved every 200 ps. We used MD Analysis<sup>112</sup> to extract the root-mean-square-fluctuation (RMSF) by aligning to the first frame of the production run and using only Ca atoms. The average RMSF is calculated between the two replicas.

### Sequence alignments

Multiple protein sequence alignments for L-TGF- $\beta$  were generated using Clustal Omega,<sup>93</sup> and 2dSS (<http://genome.lcqb.upmc.fr/2dss/contact.html>).

### Retroviral production and transduction

$\beta 8$  retroviral particles were assembled in the Phoenix 293 amphotropic cell line via transfection of pBabe  $\beta 8$  using lipofectamine 3000 reagent (Invitrogen) per the manufacturer's instructions. To maximize transfection efficiency phoenix cells were plated onto Poly-L-Lysine (Sigma Aldrich) coated 6-well culture plates at a density of  $3.5 \times 10^5$  cells/ml 1 day before transfection. Media was replaced 3 hrs prior to transfection, and retroviral supes were harvested 48 hrs after transfection. MFB-F11 cells were then transduced with retroviral supernatants by incubating  $1 \times 10^5$  MFB-F11 cells with 100  $\mu\text{L}$  of retroviral supernatant for 5 mins in suspension before plating out onto 24 well cell culture plates containing 500  $\mu\text{L}$  basal media. Puromycin resistant colonies were then selected by supplementation of basal media with 5  $\mu\text{g/mL}$  puromycin dihydrochloride (Sigma-Aldrich, #P8833). Cells were screened for the presence of cell surface  $\beta 8$  by flow cytometry.

### Flow cytometry and immune cell analysis

To normalize expression levels of cell surface L-TGF- $\beta$  proteins, TMLC cell lines were stained with anti-HA (clone 5E11D8) and green fluorescent protein (GFP) expression as a surrogate marker for L-TGF- $\beta$  expression. Cells were stained in MACS staining buffer on ice



for 15 min, before washing and staining with anti-mouse-APC conjugated secondary antibody (Biolegend # 405308). Cells were then sorted for uniform cell surface expression. To confirm the expression of human  $\beta 8$ , MFB-F11 cells were stained with anti- $\beta 8$  antibody (Clone F12) for 15 mins on ice at a concentration (1  $\mu\text{g}/\text{ml}$ ). Cells were then washed and subsequently stained with anti-mouse-APC conjugated secondary antibody (Biolegend #405308).

To measure Treg frequency in mice, peripheral blood and splenocyte cell preparations were collected. Briefly, whole blood was collected into EDTA-containing tubes, centrifuged at 700xg for 20 minutes and the subsequent cell pellet lysed with ACK lysis buffer for 10 min and washed twice with >5x volumes of staining buffer (PBS + 2.0% BSA + 0.5 mM EDTA). Splenocytes were manually dissociated, passed through a 70  $\mu\text{m}$  filter, and red blood cells lysed in ACK buffer for 5 min before washing and resuspending in staining buffer. Cells were stained with viability dye for 20 minutes at room temperature followed by blocking Fc receptors with anti-CD16/CD32 for 10 min at 4°C. Cells were then stained with an antibody cocktail diluted in Brilliant Staining Buffer (BD biosciences) with anti-mouse surface antibodies (see [key resources table](#)) for 15 min at 4°C. Cells were fixed and permeabilized using the FoxP3/Transcription Factor Fixation/Permeabilization kit (eBioscience) for 45 min at 4°C. Cells were stained overnight with intracellular antibodies, washed and ultimately resuspended in staining buffer before acquiring data the next day on an Aurora spectral flow cytometer (Cytek) with SpectroFlo (Cytek) software. Data was analyzed using SpectroFlow (Cytek) for pre-processing of spill-over matrices and FlowJo (BD biosciences) for population gating. CD4<sup>+</sup> Treg were defined by forward scatter (FSC) x side scatter (SSC) profile (lymphocyte gate), single cells, live, CD45<sup>+</sup>, Ly6g<sup>-</sup>, CD90<sup>+</sup>, CD19<sup>-</sup>, TCR $\beta$ <sup>+</sup>, CD4<sup>+</sup>, CD8a<sup>-</sup>, CD25<sup>+</sup>, FoxP3<sup>+</sup>.

### Assays to measure TGF- $\beta 1$ activation by different forms of $\alpha\text{v}\beta 8$

TMLC reporter cells expressing GARP/L-TGF- $\beta 1$  were cultured in the presence of plate-bound (immobilized)  $\alpha\text{v}\beta 8$  ectodomain, soluble  $\alpha\text{v}\beta 8$  ectodomain, soluble  $\alpha\text{v}\beta 8\text{fl-nd}$  or empty (Control) nanodisc. All integrins were plated using a 2-fold dilution series ranging from 10  $\mu\text{g}/\text{mL}$ –0.039  $\mu\text{g}/\text{mL}$ .  $\alpha\text{v}\beta 8$  ectodomain or BSA control were incubated on 96 well culture plates in the presence of PBS containing 1 mM  $\text{MgCl}_2$  and 1 mM  $\text{CaCl}_2$  for 1 hr at 37 °C. Wells were then washed 2x in basal media before  $\alpha\text{v}\beta 8$  coated wells and all other assay wells were blocked for 1 hr in basal media at 37 °C. After blocking, soluble  $\alpha\text{v}\beta 8$  ectodomain,  $\alpha\text{v}\beta 8\text{fl-nd}$  or empty nanodisc were plated in basal media using the same 2-fold dilution series described above. TMLC cells were then plated at an equal density (1x10<sup>5</sup> cells/mL in basal media) in triplicate in 96-well cell culture plates (Corning, # 3599). After attachment cells were incubated for 16 hours under standard cell culture conditions (37 °C humidified incubator, 5%  $\text{CO}_2$ ). After 16 hrs cells were lysed and assessed for luciferase activity as previously described.<sup>27,95</sup>

### Assays to measure TGF- $\beta 3$ activation

The  $\alpha\text{v}\beta 8$  ectodomain was coated along with the negative control, BSA (Sigma-Aldrich) (1  $\mu\text{g}/\text{ml}$ ) onto 96 well cell culture plates (Corning) in PBS (1mM  $\text{Ca}^{2+}$  and 1mM  $\text{Mg}^{2+}$ ) for 1 hour at RT as described.<sup>22</sup> Wells were then washed 1x in PBS and blocked in 1% BSA in PBS for 1 hr at RT. TMLC L-TGF- $\beta 1$ /GARP, TMLC L-TGF- $\beta 3$ /GARP or TMLC L-TGF- $\beta 1$ \_lasso3\_chimera/GARP were then plated onto coated wells at density of 1x10<sup>5</sup> cells/mL. After attachment cells were incubated for 16 hours under standard cell culture conditions (37 °C humidified incubator, 5%  $\text{CO}_2$ ). To measure diffusible TGF- $\beta$ , media was removed from culture wells and plated onto fresh wells containing 1.5x10<sup>4</sup> TMLC WT cells/mL for a further 16 hrs before lysis in luciferase assay lysis buffer and measurement of luciferase activity as described above. To measure cell intrinsic flexibility of L-TGF- $\beta 3$ , TMLC L-TGF- $\beta 1$ /GARP, TMLC L-TGF- $\beta 3$ /GARP or TMLC L-TGF- $\beta 1$ \_lasso3\_chimera/GARP cells were lysed and assayed for luciferase activity as above.

### Assays to measure the signaling direction of the $\alpha\text{v}\beta 8$ /L-TGF- $\beta 1$ /GARP complex

MFB-F11 TGF-  $\beta$  reporter cells expressing human  $\beta 8$  were co-cultured with TMLC TGF- $\beta$  reporter cells expressing cell surface L-TGF- $\beta 1$  via the adaptor molecule GARP. Cells were plated at an equal density (1x10<sup>5</sup> cells/mL in basal media) in triplicate in 96-well cell culture plates (Corning, # 3599). To quantify TGF- $\beta$  signaling, TMLC or MFB-F11 cells were also plated alone at a density of 1x10<sup>5</sup> cells/mL in basal media supplemented with known concentrations of rhTGF- $\beta 1$  (R&D systems, #240-B-002/CF). After attachment cells were incubated for 16 hours under standard cell culture conditions (37°C humidified incubator, 5%  $\text{CO}_2$ ). After 16 hours conditioned media was removed for assessment of SEAP activity. Attached cells were washed once in PBS and subsequently lysed in an equal volume of luciferase assay lysis buffer for 30 mins at room temperature to detect luciferase activity (Biotium, # 26140-079). Measurement of SEAP activity was achieved using the SEAP reporter gene assay kit (Abcam, #133077). Briefly, conditioned media was heated at 65 °C to denature endogenous alkaline phosphatases. Conditioned media samples were then transferred to 96-well cell culture white plates (Costar, # 3917) and an equal volume of SEAP substrate was added to each well. Samples were incubated for 5 min before Luminescence was recorded using a luminescence assay plate reader (Promega: Glomax explorer). Measurement of luciferase activity was performed using a luciferase assay kit as described above. SEAP or luciferase activity was converted to ng/ml TGF- $\beta$  signaling by interpolation from the standard curve of known rhTGF- $\beta 1$  treatments. These assays were repeated with the following cell lines: TMLC L-TGF- $\beta 1$  (RGD)/GARP, TMLC L-TGF- $\beta 1$  (RGE)/GARP and TMLC L-TGF- $\beta 3$ /GARP.

### TGF $\beta$ R2 binding assay

ELISA plates were coated with 10  $\mu\text{g}/\text{ml}$  recombinant  $\alpha\text{v}\beta 8$  in 20 mM HEPES pH 8.0, 150 mM NaCl (HBS) with 1mM  $\text{CaCl}_2$  and 1 mM  $\text{MgCl}_2$  for 1 hr at RT. Wells were then washed in HBST (0.5% Tween-20) and blocked in HBS with 2.5% BSA for 1 hr at RT. Wells were

then washed in HBS-T (0.5% Tween-20) and applied by 8  $\mu\text{g/ml}$  TGF- $\beta$ 1/GARP, 4  $\mu\text{g/ml}$  TGF- $\beta$ 3/GARP or in HBS with 1 mM  $\text{CaCl}_2$  and 1 mM  $\text{MgCl}_2$ , after 5 minutes incubation at RT, serial dilutions of recombinant TGF- $\beta$ 2-Fc in HBS with 1 mM  $\text{CaCl}_2$  and 1 mM  $\text{MgCl}_2$  were added for 1 hr at RT. Wells were then washed in HBST (0.5% Tween-20) and antibody were added (2  $\mu\text{g/ml}$ ) in HBS with 1 mM  $\text{CaCl}_2$  and 1 mM  $\text{MgCl}_2$  for 1 hr at RT. After washing in in HBST (0.5% Tween-20), bound antibodies were detected using anti-mouse-HRP using TMB substrate and colorimetric detection (Glomax Explorer, Promega).

### iTreg conversion assay

CD4<sup>+</sup> T cells isolated from whole mouse spleen using a CD4<sup>+</sup> T cell isolation kit (Miltenyi, Cat:130-104-454 as previously described<sup>27</sup>). Purified CD4<sup>+</sup> T-cells were then cultured in Serum Free Medium, containing IMDM (Gibco, Cat:12440-053), 1% Insulin-Transferrin-Selenium (Gibco, Cat:41400-045), 50  $\mu\text{M}$  2-Mercaptoethanol (Gibco, Cat:31350-010), 2mM Retinoic acid (Sigma, Cat:R2625), and 135 U/ml IL-2 (R&D, Cat:402-ML/CF) for 72 h in 24-well plates with at a density of  $1.0 \times 10^6/\text{ml}$ . Cells were plated onto wells coated with 6  $\mu\text{g/mL}$  anti-Mouse-CD3 (Biolegend, Cat:10020) with 12  $\mu\text{g/mL}$  recombinant  $\alpha\text{v}\beta$ 8 ectodomain or BSA control. Recombinant human TGF- $\beta$ 1 (1 ng/mL, R&D, Cat:240-B) was also used as a positive control. After 72 h, the cells were immunostained with anti-mouse-CD25-APC (Biolegend, Cat:102012) and anti-mouse-Foxp3-PE (Biolegend, Cat:126404) and detected by flow cytometry.

### Immunoblots

Cells were lysed with RIPA buffer (Sigma, Cat:R0278) supplemented with protease inhibitor cocktail (Thermo Scientific<sup>TM</sup>, Cat:87786) and phosphatase inhibitor (Thermo Scientific<sup>TM</sup>, Cat:A32957). Protein concentration was determined by BCA assay (Thermo Scientific<sup>TM</sup>, Cat:23228). The precast gels (Cat:456-1084) and transfer pack (Cat:1704158) were bought from Bio-Rad. SDS-PAGE and WB were performed according to the manufacturer's instructions. Anti-TGF- $\beta$ 1 (Cat:ab179695), anti-p-Smad 2/3 (CST, Cat:8828S), anti-Smad 2/3 (CST, Cat:3102S), anti-Na.K-ATPase (Invitrogen, Cat:MA5-32184), and anti-Actin (Sigma, Cat:A2228-200ul) were all used with a dilution factor of 1:1000. anti-Mouse-IgG (Jackson ImmunoResearch, cat:711-035-152) and anti-Rabbit-IgG (Jackson ImmunoResearch, cat:715-035-150) were used with a dilution factor of 1:5000. When assessing protein expression of TGF- $\beta$ 1 in mouse plasma, it was necessary to pre-clear endogenous IgGs using 3 sequential overnight Sepharose G incubations at 4 C. The supernatant was then used for WB detection of TGF- $\beta$ 1 expression levels in plasma.

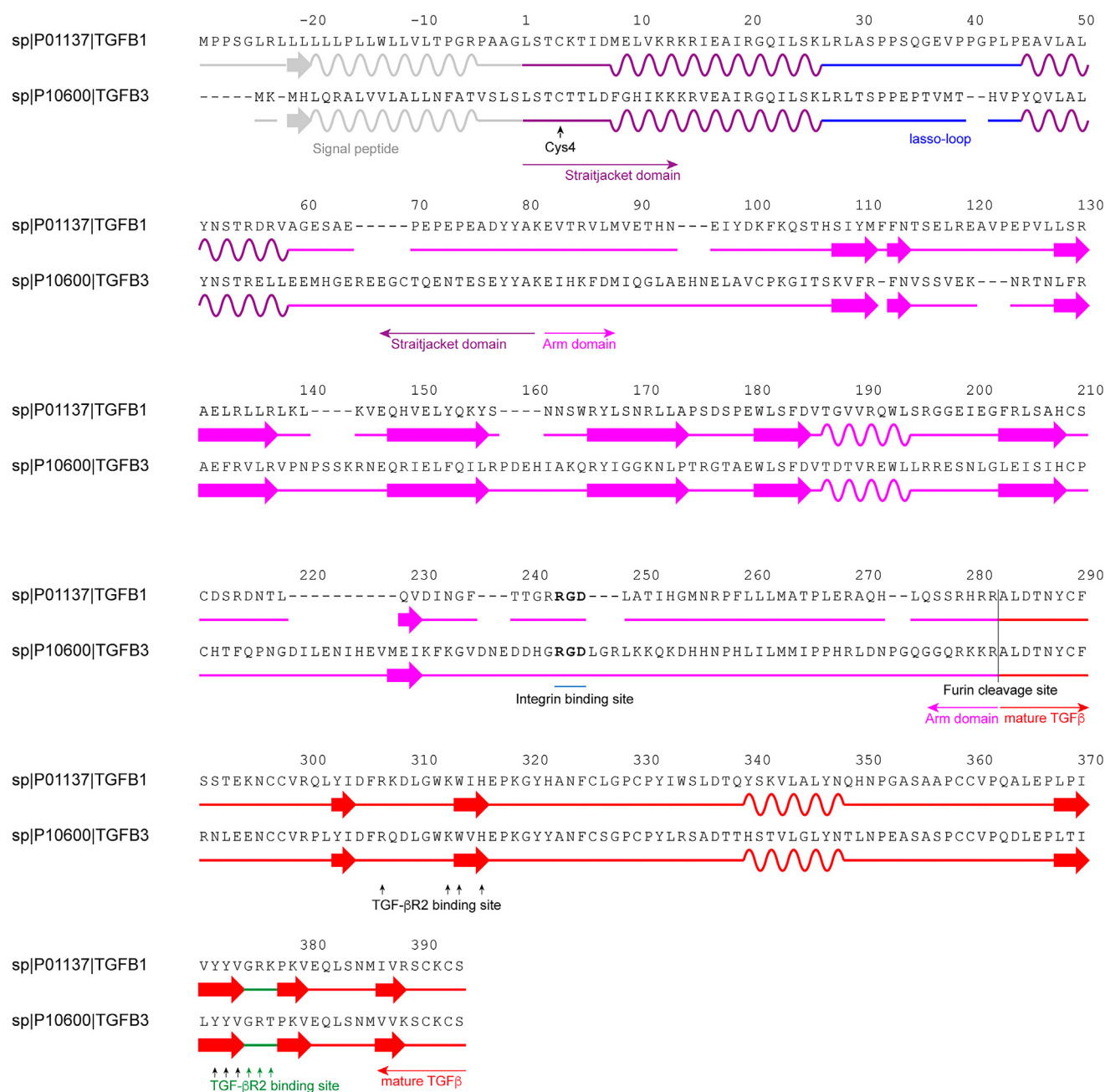
### Cell surface biotinylation

WT CD4<sup>+</sup> T cells were harvested after activation, as above, from culture wells by gentle pipetting. After 3 washes with cold PBS, cells were resuspended with ice-cold PBS at a density of  $2.5 \times 10^7/\text{ml}$ . NHS-LC-Biotin was (Thermo Scientific<sup>TM</sup>, cat: A39257) was used following the manufacturer's instructions. WT CD4<sup>+</sup> T-cells were lysed before or after surface biotinylation. Lysate from non-biotinylated control cells were compared to eluates from equal amounts of biotinylated and non-biotinylated lysates applied to streptavidin agarose (SA), with anti-mature TGF- $\beta$ 1. A band the same size of cleaved mature TGF- $\beta$ 1 was seen in eluates from SA beads incubated with non-biotinylated lysate, making quantitative comparisons of cleaved to uncleaved forms of TGF- $\beta$ 1 difficult. Membranes were stripped and reprobed with anti-Na<sup>+</sup>/K<sup>+</sup> ATPase, as a cell membrane marker.

### QUANTIFICATION AND STATISTICAL ANALYSIS

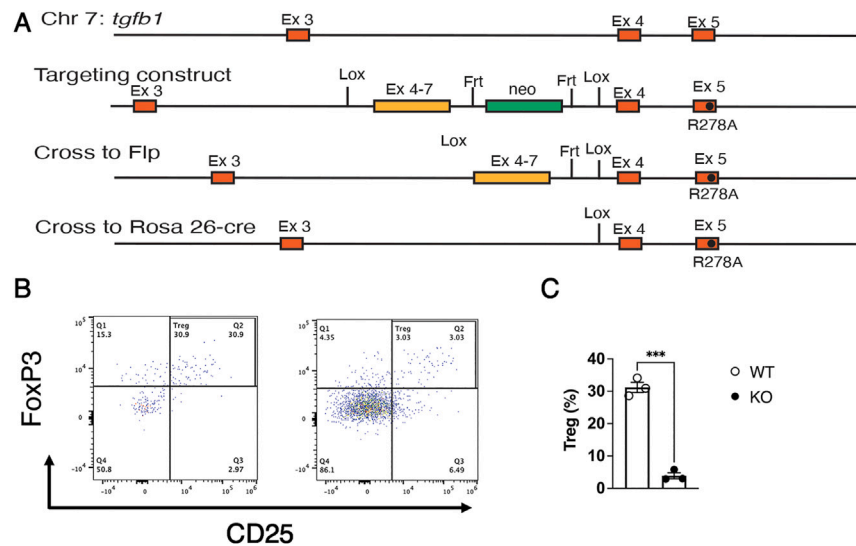
Assays are reported as means  $\pm$  s.e.m., ( $\geq 3$ ). Statistical analyses were performed using Prism 9 (GraphPad Software, San Diego, CA).

# Supplemental figures



**Figure S1. Domain organization of L-TGF-β1 and -β3, related to the introduction and Figures 2 and 4**

Sequence alignment with secondary structure prediction of L-TGF-β1 and L-TGF-β3. Domains are colored as in Figure 2A, and secondary structure prediction is marked. Sequence numbering does not include signal peptide and is consistent in all figures and text.



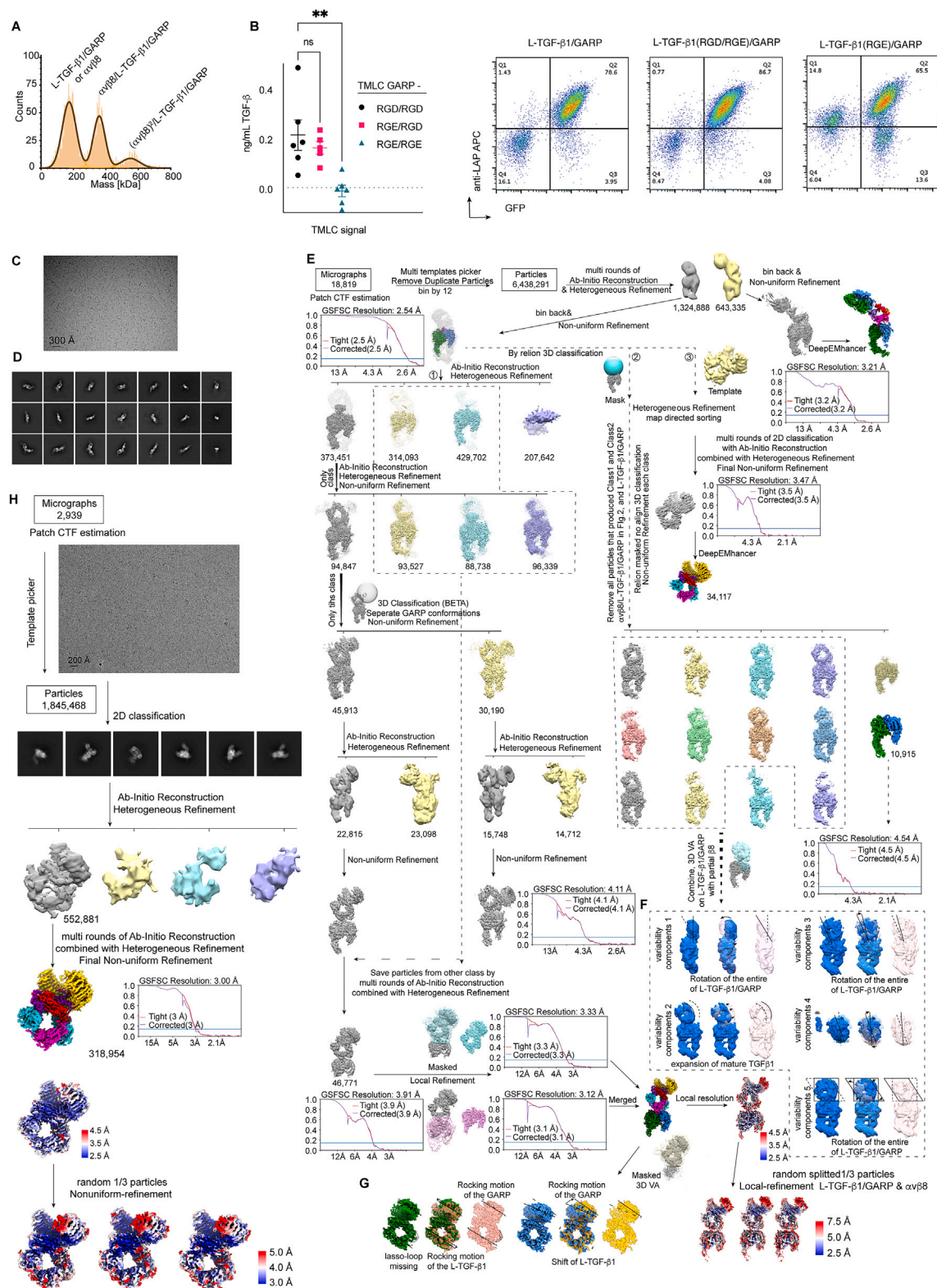
**Figure S2. Engineering of *tgfb1* R278A KI/KI mice and immune characterization of *tgfb1*<sup>-/-</sup> mice, related to Figure 1**

(A) From top to bottom, schematic of the *tgfb1* targeted locus, targeting construct after germline integration, after crossing to mice expressing flippase (flp), after crossing to rosa 26-cre.

(B) Scatter plots showing Treg from PBMC isolated from post-natal day 18 *tgfb1*<sup>-/-</sup> (KO/KO) mice compared with day 18 littermate controls (WT/KO). Staining done as in Figure 1.

(C) Scatter bar graphs showing Treg as a percentage of CD4<sup>+</sup> T cells ( $n = 3$  KO/KO;  $n = 3$  WT/KO). Shown is mean and SEM. \*\*\* $p < 0.001$ .





**Figure S3. Single-particle cryo-EM and image processing of L-TGF- $\beta$ 1/GARP and  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP complex, related to Figure 2**  
(A) Mass photometry analysis of L-TGF- $\beta$ 1/GARP mixed with ectodomain of  $\alpha$ v $\beta$ 8. Three peaks correspond to L-TGF- $\beta$ 1/GARP or  $\alpha$ v $\beta$ 8 alone at ~180 kDa, one L-TGF- $\beta$ 1/GARP bound with one  $\alpha$ v $\beta$ 8 at ~380 kDa, and one L-TGF- $\beta$ 1/GARP bound with two  $\alpha$ v $\beta$ 8 integrins at ~560 kDa.

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(B) Co-culture of MFB-F11  $\alpha\text{v}\beta 8$  expressing cells with TMLC L-TGF- $\beta 1$ /GARP presenting cells is sufficient to activate the TGF- $\beta$  signaling pathway by TMLC. To test whether one RGD site is sufficient for  $\alpha\text{v}\beta 8$ -mediated TGF- $\beta$  activation, TMLC-L-TGF- $\beta 1$ /GARP cells were made using a 1:1 transfection ratio of TGF- $\beta 1$  (RGD) and TGF- $\beta 1$  (RGE) (pink squares). TMLC: L-TGF- $\beta 1$ (RGE/RGE)/GARP cells are also included to demonstrate the requirement of RGD for  $\alpha\text{v}\beta 8$ /L-TGF- $\beta 1$  binding and activation (green triangles). The results (vertical axis) are normalized against a standard TGF- $\beta$  activation curve. Shown is mean and SEM.  $^{**}p < 0.01$  by one-way ANOVA followed by Sidak's multiple comparison test for the indicated comparisons.

(C) Representative electron micrograph of frozen-hydrated sample of mixing  $\alpha\text{v}\beta 8$  with L-TGF- $\beta 1$ /GARP in 1:1 molar ratio.

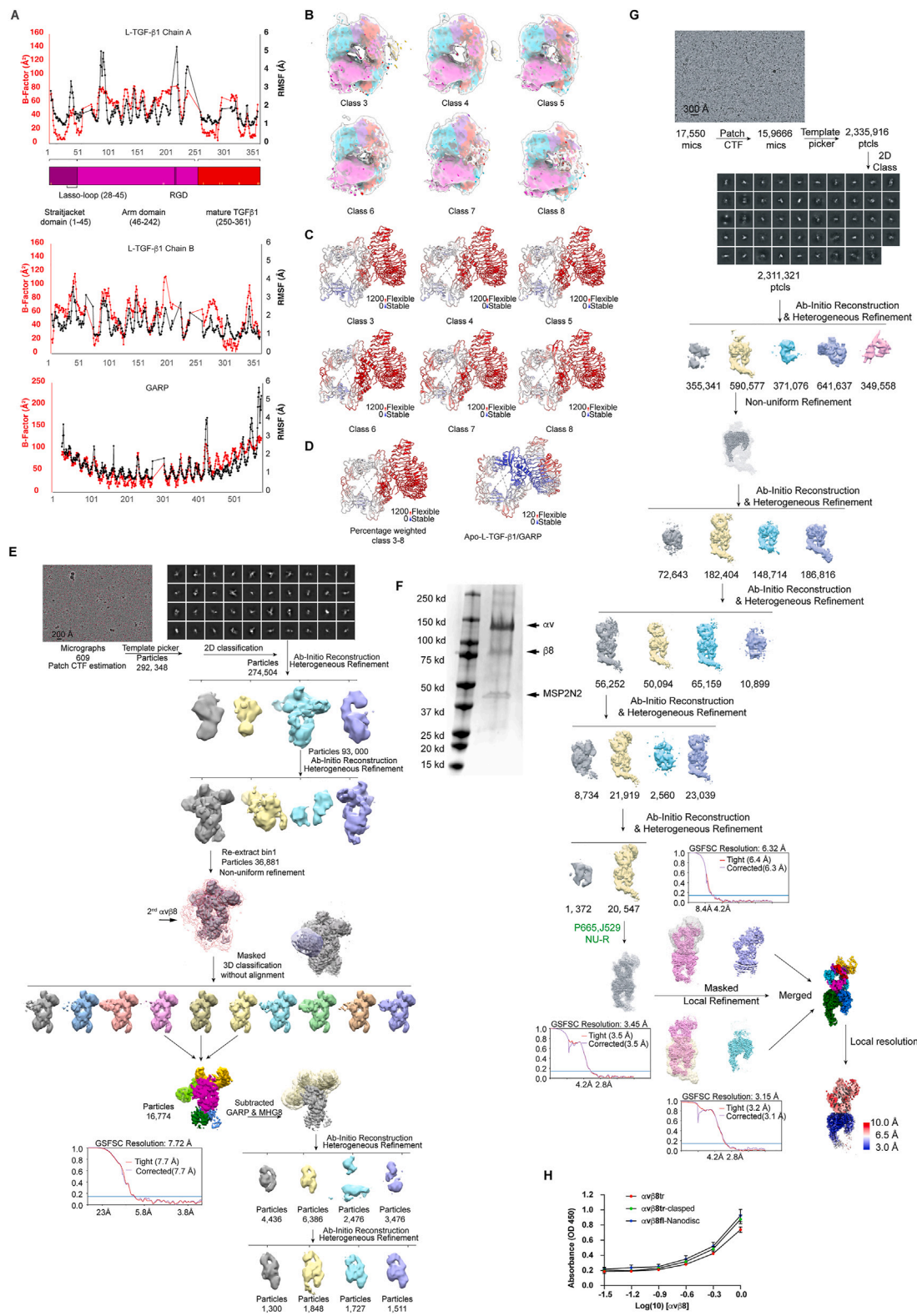
(D) Representative 2D class averages calculated from particles selected for further processing.

(E) Flow chart of cryo-EM data processing. FSC curves for resolution estimations are included at various steps of the flow chart.

(F) Motions captured in  $\alpha\text{v}\beta 8$ /L-TGF- $\beta 1$ /GARP complex along five eigenvectors from 3DVA are illustrated.

(G) 3DVA of particles in class 1 in [Figure 2F](#).

(H) Image processing flow chart from representative electron micrograph to the final reconstruction and FSC for final resolution estimation of L-TGF- $\beta 1$ /GARP.

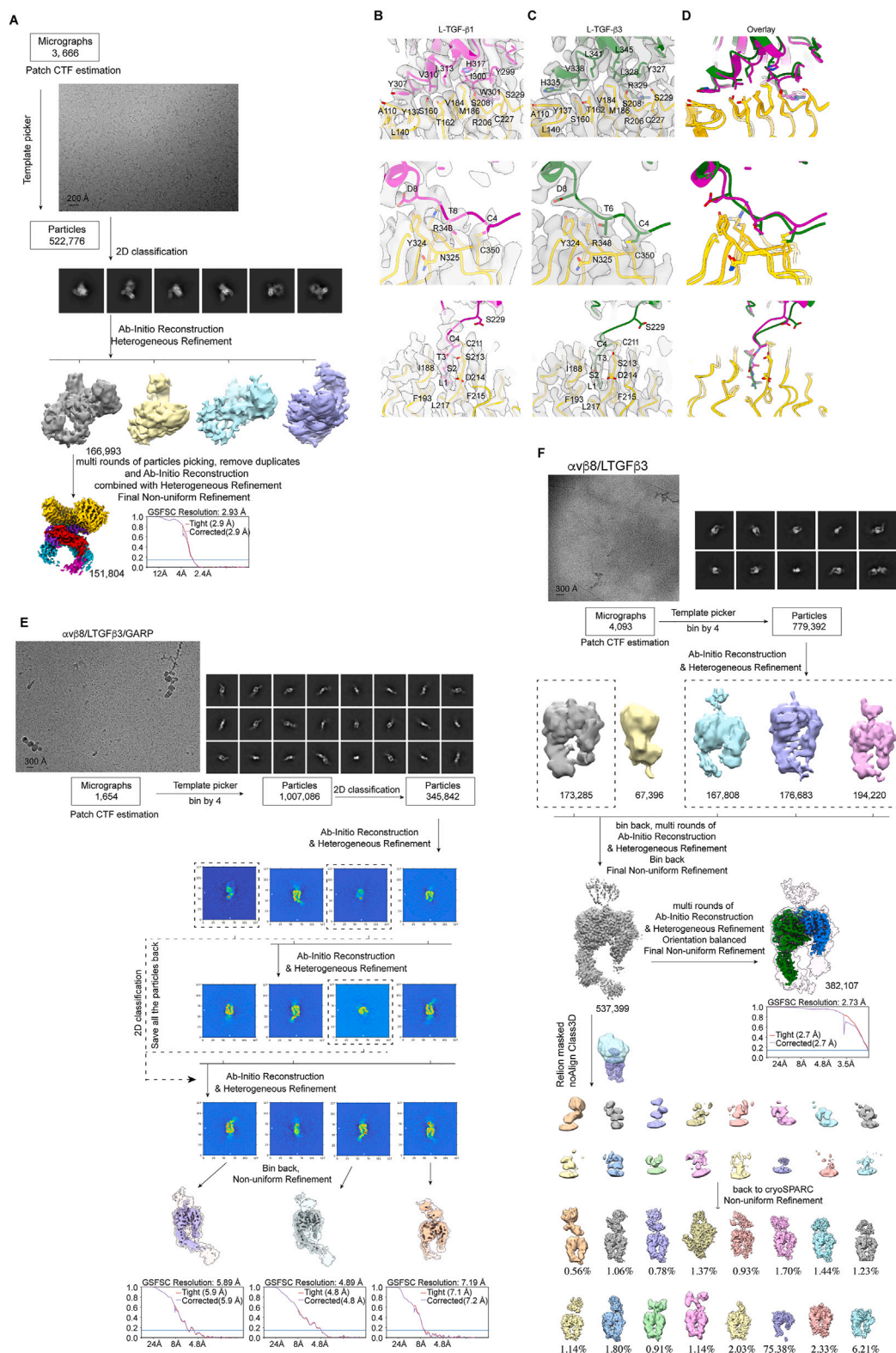


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**Figure S4. Single-particle cryo-EM, molecular dynamics of L-TGF- $\beta$ 1/GARP, and image processing of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP bound with Fab MHG8, related to Figure 3**

- (A) The B-factor from cryo-EM analysis (red) and the average RMSF (black) from long molecular dynamics simulations are shown for each residue of L-TGF- $\beta$ 1/GARP where flexible residues correlate with high B-factors.
- (B) The local sharpen maps (colored) superimposed in related low passed 8 Å maps (gray in transparent) of class 3–8 from Figure 2F.
- (C) B-factor analysis of models for (B).
- (D) The weighted averaged B-factor of the models in (B) (left) compared with apo L-TGF- $\beta$ 1/GARP (right).
- (E) Image processing flow chart from representative electron micrograph to the final reconstruction and FSC for final resolution estimation of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 1/GARP bound with Fab MHG8.
- (F) SDS-PAGE of  $\alpha$ v $\beta$ 8fl-nd.
- (G) Image processing flow chart from representative electron micrograph to the final reconstruction and FSC for final resolution estimation of  $\alpha$ v $\beta$ 8fl-nd/L-TGF- $\beta$ 1/GARP.
- (H) The binding affinity assay of L-TGF- $\beta$ 1/GARP with  $\alpha$ v $\beta$ 8tr,  $\alpha$ v $\beta$ 8tr-clasped, and  $\alpha$ v $\beta$ 8fl-nd showing no difference.



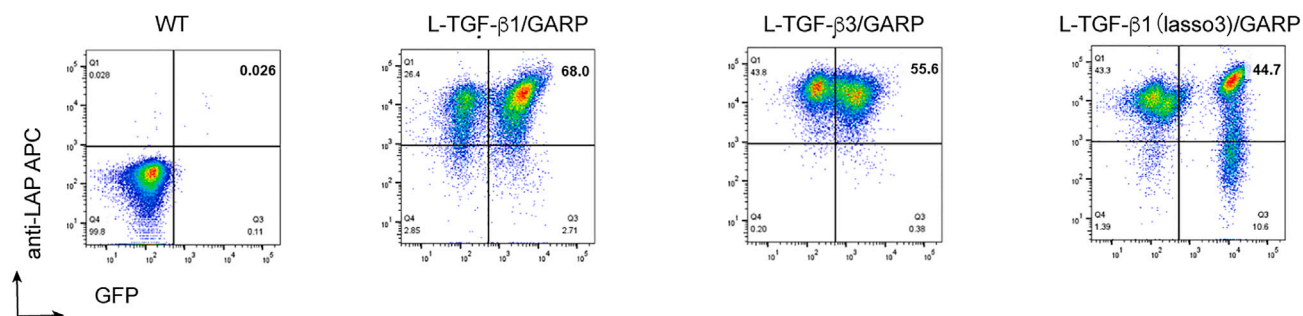


**Figure S5. Single-particle cryo-EM and image processing of L-TGF-β3/GARP, αvβ8/L-TGF-β3/GARP complex and comparison of binding interface between GARP and L-TGF-β, related to Figure 4**

(A) Image processing flow chart from representative electron micrograph to the final reconstruction and FSC for final resolution estimation for L-TGF-β3/GARP.

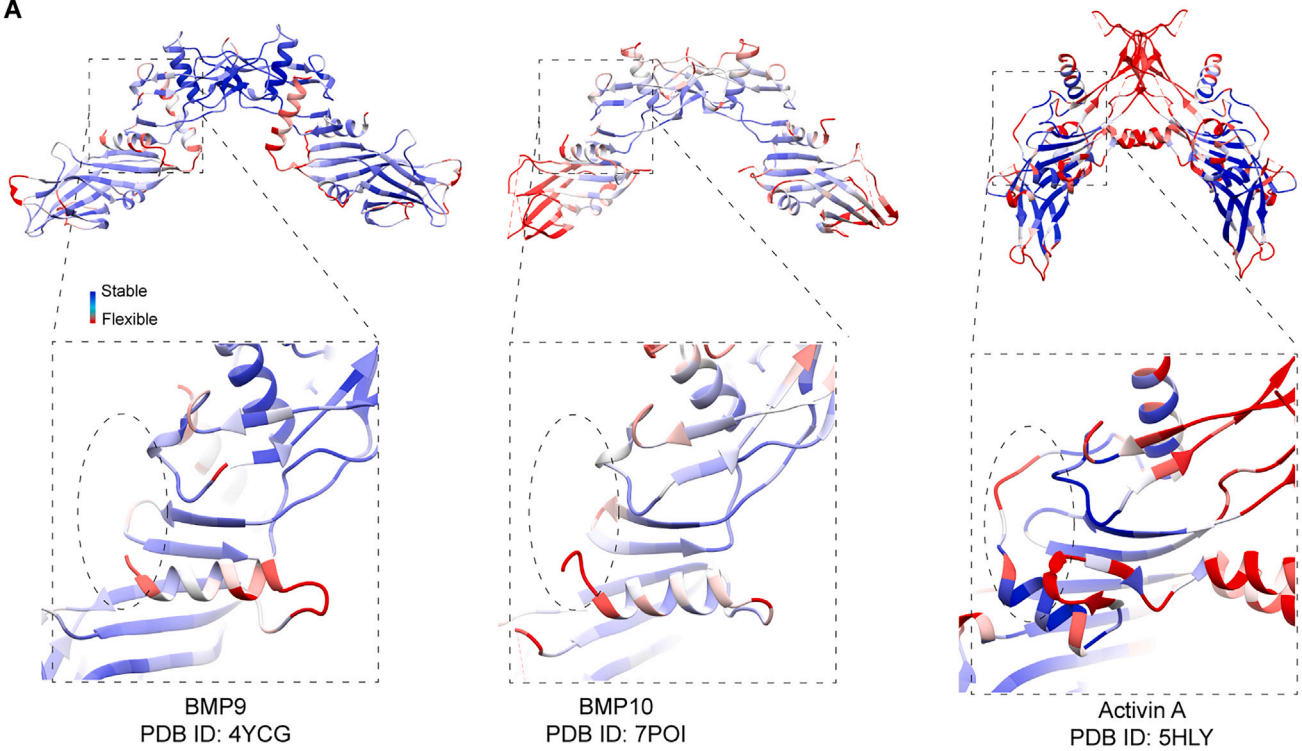
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- (B) Three regions of cryo-EM density map and docked atomic model selected from the interface between L-TGF- $\beta$ 1 (magenta) and GARP (yellow).
- (C) Three regions of cryo-EM density map and docked atomic model selected from the interface between L-TGF- $\beta$ 3 (green) and GARP (yellow).
- (D) Overlay of the atomic models of L-TGF- $\beta$ 1/GARP (magenta/yellow) and L-TGF- $\beta$ 3/GARP (green/yellow) from the same selected regions.
- (E) Flow chart of processing single-particle cryo-EM dataset of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3/GARP complex.
- (F) Flow chart of processing single-particle cryo-EM dataset of  $\alpha$ v $\beta$ 8/L-TGF- $\beta$ 3 complex.



**Figure S6.** Representative flow cytometry scatter plots demonstrating the percentage of cell surface L-TGF- $\beta$ 1, L-TGF- $\beta$ 3, and L-TGF- $\beta$ 1 (lasso3), respectively, related to [Figure 5](#)

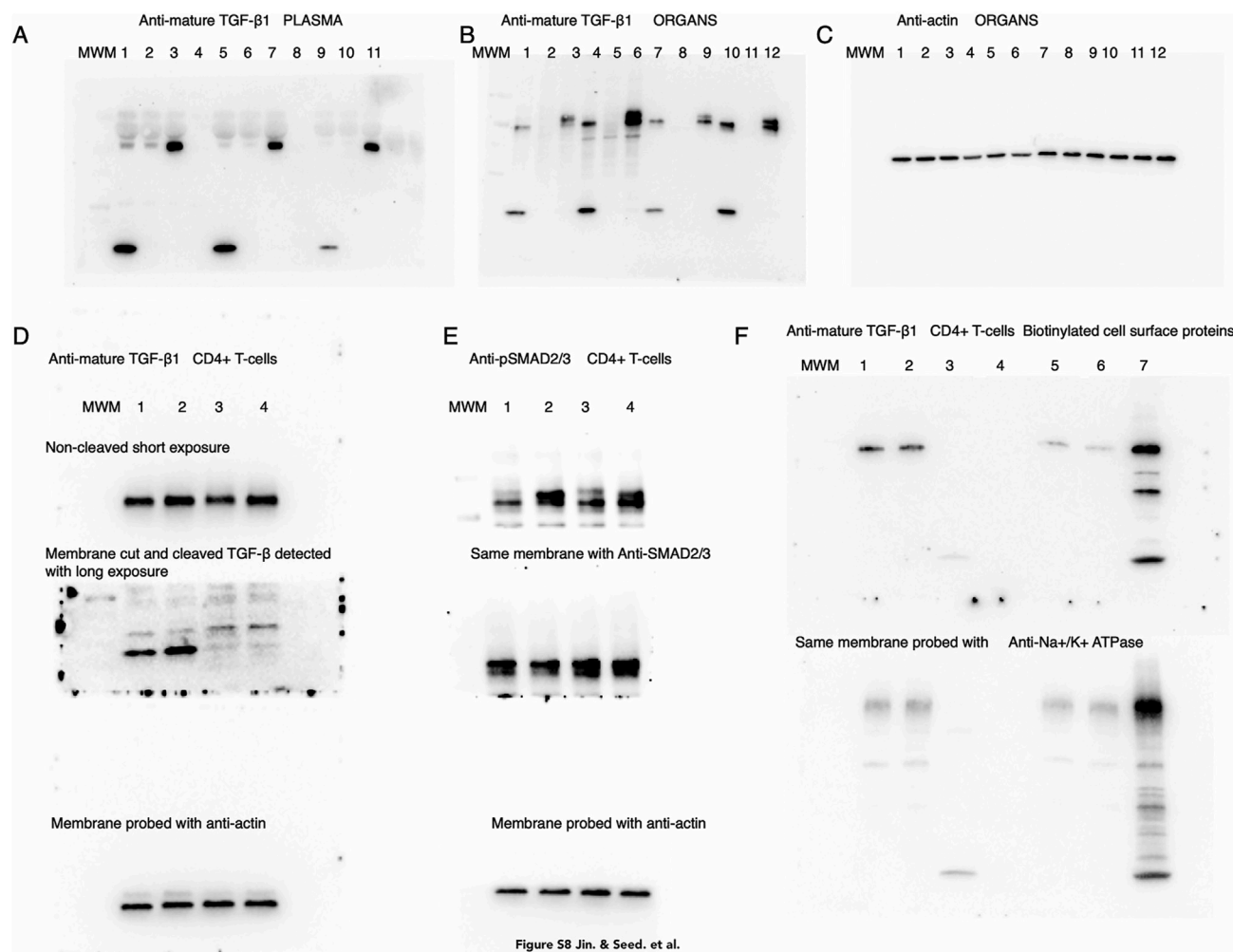
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**Figure S7. B-factor for each pro-TGF- $\beta$  superfamily crystal structure, related to the discussion and Figure 5**

Ribbon diagram of BMP9 (PDB: 4YCG), BMP10 (PDB: 7POI), and activin A (PDB: 5HLY). Residues are colored by B-factors. Residues with lower B-factor (blue) are relatively stable, while those with higher B-factor (red) are more flexible. An enlarged view within dashed box for each shows that the area around TGF- $\beta$ R binding site, in which each lasso loop (dashed oval) is flexible.





**Figure S8. Raw immunoblots, related to Figure 1**

(A) Raw immunoblot of plasma data depicted in Figure 1M. Plasma was subject to 3 sequential rounds of IgG pre-clearance using Sepharose G beads. Lanes 9 (WT/WT), 10 (KO/KO), and 11 (KI/KI) are depicted in Figure 1M. Samples in lanes 1, 2, and 3 and 5, 6, and 7 represent the samples after 1 or 2 rounds of pre-clearing, respectively.

(B) Raw immunoblot of TGF- $\beta$ 1 expression in murine organ homogenates (Figure 1M). Lanes 1, 2, and 3 show TGF- $\beta$ 1 staining from WT/WT, KO/KO, and KI/KI mice, respectively, in kidney; lanes 4, 5, and 6 in liver; lanes 7, 8, and 9 in lung; and lanes 10, 11, and 12 in spleen.

(C) Raw immunoblot representing the  $\beta$ -actin expression depicted in Figure 1M as a loading control.

(D) Raw immunoblot data for Figure 1N demonstrating the expression of uncleaved TGF- $\beta$ 1 (upper blot) after 0.9 s exposure and cleaved TGF- $\beta$ 1 (lower blot) after 14.9 s exposure. Included below the TGF- $\beta$ 1 blots is a raw immunoblot stained with anti- $\beta$ -actin to ensure equal protein loading.

(E) Raw immunoblot of anti-pSMAD2/3 (upper blot) and anti-SMAD2/3 (lower blot) presented in Figure 1P. Included below the SMAD2/3 blot is an immunoblot using anti- $\beta$ -actin to ensure equal protein loading.

(F) Raw immunoblots for data depicted in Figure 1O. The upper blot displays anti-TGF- $\beta$ 1 staining. Lane 1 (non-biotinylated lysate), lane 2 (non-biotinylated lysate after Sepharose pre-clearance), lane 3 (eluate from pre-cleared lysate incubated with streptavidin beads), lane 4 (blank), lane 5 (biotinylated lysate), lane 6 (biotinylated lysate after Sepharose pre-clearance), lane 7 (eluate from pre-cleared biotinylated lysate incubated with streptavidin beads). The lower image depicts the raw immunoblot data depicting anti-Na<sup>+</sup>/K<sup>+</sup> ATPase staining to demonstrate cell surface protein enrichment. Figure 1O depicts lane 5, lane 7, and lane 3, respectively.